From Returns to Defensible Portfolio Evidence

Graduate Lecture Notes in Uncertainty-Aware Portfolio Benchmarking

Professor-Level Teaching Edition for a BSc Mathematics Graduate

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Contents

Chapter 1

Orientation: what you will be able to do

By the end, a student should be able to reproduce every analytical task in the paper, explain why it is needed, distinguish a valid claim from an overclaim, and design a leakage-free portfolio benchmarking pipeline from raw monthly data to an auditable report. This orientation chapter sets the scope, states measurable learning outcomes, previews the data and the headline results, maps the chapters, lists the prerequisites and conventions, and warns of the misconceptions the course is built to prevent. Read it once before starting and return to the roadmap and the misconception table whenever a later chapter feels disconnected from the whole.

1.1 What these notes are about

These notes teach a single, self-contained empirical study end to end: the risk-adjusted benchmarking of the 25 Fama–French size and book-to-market portfolios. The object of study is deliberately modest — 25 passively constructed, value-weighted characteristic portfolios and three factor return series — because the difficulty lies not in the data but in *what may honestly be concluded from it.* The same 25 portfolios are the classical test assets from which the size and value factors were themselves built, so their factor "alpha" is a model-relative residual, not a measure of skill.

The intellectual arc is one long argument. We first **describe** the historical cross-section of alpha; then **quantify** how much of the apparent ordering could be sampling noise; then test whether the ordering **survives** resampling, time and a change of benchmark; and finally ask whether a ranking formed from the past carries genuine **predictive** content for the future. The punchline, developed carefully across the chapters, is a distinction that is easy to state and easy to get wrong: a *stable* cross-sectional ordering does carry out-of-sample information, but *re-estimating* that ordering through time — the thing practitioners usually call "dynamic" skill — adds nothing here. Learning to see, defend and not overstate that distinction is the point of the course.

Two commitments run through every chapter. The first is **uncertainty-first**: a number is never reported without an honest statement of how much it could move under resampling, model choice or time. The second is **reproducibility**: every figure, table and claim is regenerated by a deterministic pipeline from hashed inputs, so a reader can audit rather than trust.

1.2 Learning outcomes

On completing these notes a student should be able to:

1. **Estimate.** Fit a factor regression for each portfolio by ordinary least squares, and explain precisely why the intercept is called alpha and what it does *not* measure.
2. **Quantify dependence.** Build the full 25 × 25 heteroskedasticity- and autocorrelation-consistent (HAC) covariance of the alpha vector, and explain why 25 separate standard errors are insufficient when the portfolios share dates and shocks.
3. **Pool.** Derive and apply multivariate empirical-Bayes shrinkage, and articulate the winner's-curse problem it addresses.
4. **Test.** Carry out the joint HAC/Wald test and the Gibbons–Ross–Shanken test, control the false-discovery rate across 25 hypotheses, and say which test is primary and why.
5. **Resample.** Implement a common-date circular block bootstrap that refits every stage, and read a rank interval correctly as a resampling distribution rather than a Gaussian confidence interval.
6. **Validate forward.** Construct a strictly leakage-free forward evaluation with frozen training estimates, and prove that no future information enters a score, rank or loading.
7. **Decompose.** Run the incremental test that separates the dynamic content of a rolling ranking from the static content of a fixed ordering, and recognise it as a Diebold–Mariano test of equal predictive accuracy.
8. **Interpret.** Separate a supported claim from an overclaim, and write a conclusion that survives adversarial reading.

The verbs are deliberate: the assessment (Chapter 18) asks a student to *do* each of these, not merely to recognise them.

1.3 The dataset at a glance

Quantity	Value in these notes
Cross-section N	25 size × book-to-market portfolios
Months T	1,193 consecutive months, July 1926 – November 2025
Panel size	29,825 portfolio-months, balanced, no calendar gaps
Benchmark factors K	3 for FF3 (Mkt−RF, SMB, HML); 5 for the FF5 sensitivity
Return units	Decimal monthly excess return, $y_{it} = r_{it} - r_{ft}$
Annualisation	Monthly decimal alpha × 12 × 10,000 = annualised basis points
Bootstrap	5,000 circular 12-month blocks, seed 2026
Rolling design	120-month training windows, stepped 12 months; 12-month forward horizon

The panel is validated by a *fail-closed* loader (Chapter 4): it raises rather than proceeds if it finds a duplicate portfolio-month key, a null required field, within-month factor disagreement, an unbalanced panel or a missing calendar month. Data integrity is treated as part of the statistical specification, not as pre-processing hygiene.

1.4 Headline results you will be able to defend

The following numbers are the destination. Each is derived, audited and qualified in a later chapter; they are collected here so that the machinery always has a visible purpose.

Finding	Figure
Joint zero-alpha hypothesis rejected (HAC/Wald)	$p = 6.4 \times 10^{-10}$; GRS $p = 1.2 \times 10^{-7}$ (secondary)
Best and worst posterior alpha	`SMALL HiBM` +132; `SMALL LoBM` −185 bps/year
Individually significant alphas after FDR	5 of 25 survive 5% Benjamini–Hochberg
Point ranking is not a precise hierarchy	bootstrap rank intervals wide; only 2 of 25 posterior intervals exclude zero
Long-horizon mobility	stay-in-quintile probability ≈ 25% over 120 months (near the 20% of pure mobility)
Forward quintile spread (top minus bottom)	243 bps/year, HAC 95% CI [153, 333], positive in 73% of windows
Incremental (rolling minus expanding) skill	−0.042, HAC $t = -1.87$: *no* dynamic edge
Benchmark dependence	`ME5 BM2` moves rank 9 (FF3) to rank 20 (FF5)
Stochastic-frontier support	only 1 of 25 boundary tests survives FDR
Data provenance	local vs. current official snapshot differ; classified NOT_COMPARABLE, not PASS

The single most important line is the incremental result: the positive forward spread is real, but it is explained by a fixed, persistent characteristic pattern, not by dynamic timing.

1.5 How the chapters fit together

Ch.	Topic	What you should be able to do afterwards
1	Orientation	State the argument, the data and the conventions
2	Preliminaries: mathematical foundations	Derive every tool used later — OLS, HAC, empirical Bayes, the bootstrap, Diebold–Mariano, and more
3	The research question	Frame the estimand and its seven difficulties
4	Data, construction, units, provenance	Build a validated balanced panel and separate replication from reconciliation
5	Factor regression and the meaning of alpha	Derive OLS alpha and interpret it as model-relative
6	Full cross-portfolio Newey–West HAC	Assemble and audit the joint 25×25 covariance
7	Joint testing and false discovery	Run HAC/Wald and GRS; control the FDR across 25 tests
8	Multivariate empirical Bayes	Derive and apply partial pooling
9	Circular block bootstrap	Produce rank and score intervals by full refit
10	Rolling persistence and transitions	Diagnose the overlap illusion; read mobility
11	Strictly forward validation	Freeze training estimates and prove no leakage
12	The incremental test	Separate dynamic forecast from stable ordering
13	Residual diagnostics and model risk	Justify HAC and FF3/FF5 sensitivity
14	Stochastic frontier analysis	Derive it fully and see why it stays a narrow diagnostic
15	External validation and status	Assign a defensible validation status
16	Reproducible implementation	Run the deterministic pipeline end to end

Ch.	Topic	What you should be able to do afterwards
17	Reading the paper's evidence	Map each claim to its supporting layer
18	Capstone	Reproduce the complete study to an assessment standard
19	Exercises with guided solutions	Practise every derivation and audit
20	Terminology glossary	Use each term precisely
21	References and further reading	Trace every method to a source

1.6 Prerequisites

The course is written for a numerate reader — a BSc mathematics graduate is the target — but it is self-contained wherever a finance concept is introduced. The prerequisites below are the tools assumed without re-derivation; everything specific to asset pricing is built up in the chapters.

1.6.1 Mathematics

Linear algebra: vectors and matrices, the transpose, matrix products, symmetric and positive (semi-)definite matrices, the eigen-decomposition, matrix inverses and, more importantly, why one solves a linear system rather than forming an inverse. Probability: expectation, variance and covariance as operators; the multivariate normal density and its quadratic form; conditional and marginal normals. Analysis and optimisation: gradients, first-order conditions, completing the square, and one-dimensional numerical optimisation. Asymptotics at the level of the law of large numbers and the central limit theorem, enough to read "$\hat{\theta}$ is consistent and asymptotically normal" without alarm. Each of these appears in a derivation: the eigen-decomposition floors a covariance matrix (Chapter 6), completing the square yields the empirical-Bayes posterior (Chapter 8), and quadratic forms drive the Wald statistic (Chapter 7); all are also derived in the preliminaries of Chapter 2.

1.6.2 Statistics

Ordinary least squares and its finite-sample and asymptotic properties; the meaning of a standard error as sampling variability of an estimator, which is *not* the volatility of the underlying returns; the logic of a hypothesis test and the correct reading of a p-value as a tail probability under a null, never as the probability that the null is true; confidence intervals as repeated-sampling statements; the idea of resampling (the bootstrap) to approximate a sampling distribution; and the multiple-comparisons problem, which motivates false-discovery control. A reader who has only ever run a single regression and read a single t-statistic will find the leap to *joint* inference over 25 correlated estimates the main conceptual hurdle of the course; it is worth pausing on.

1.6.3 Computing

Python 3 with NumPy, pandas, SciPy, statsmodels and matplotlib; `pytest` for tests; and a reproducible command-line workflow (virtual environments, pinned requirements, a single entry-point script, and deterministic random seeds). No deep software engineering is assumed, but the course insists on habits that make results auditable: pinned dependencies, hashed inputs, a recorded run manifest, and tests that fail loudly. The reference implementation targets CPython 3.13.

1.6.4 A five-minute self-check

If the following questions are comfortable, the prerequisites are met; if two or more are not, the cited chapter fills the gap.

1. Why does $(X'X)^{-1}X'y$ minimise $\|y - X\theta\|^2$? (Chapter 2)
2. If two estimators are positively correlated, is the variance of their difference larger or smaller than the sum of their variances? (Chapter 2)
3. A p-value is 0.03. State one thing it does and one thing it does *not* mean. (Chapter 2)
4. Why does choosing the largest of 25 noisy estimates bias it upward? (Chapter 2)
5. Why can a correlation between a past ranking and future returns exist even with no dynamic forecasting skill? (Chapters 10 and 12)

1.7 Mathematical conventions used throughout

Notation	Meaning	Example
Lower-case italic	Scalar	r_{it}, α_{i}, σ
Bold or context-defined lower-case	Vector	$\mathrm{f_t}$ is K × 1; α is N × 1
Upper-case	Matrix or sample size	X is T × (K+1); V is N × N
Prime, $'$	Transpose	X$'$X
Hat, $\widehat{}$	Estimated quantity	$\hat{\alpha}$, $\hat{\beta}$, $\hat{V}$
E(·), Var(·), Cov(·)	Expectation, variance, covariance	Var($\hat{\alpha}$)
N, T, K	Portfolios, months, factors	N=25, T=1,193, K=3 for FF3
y_{it}	Excess return of portfolio i in month t	$r_{it} - r_{ft}$
f_t	Factor return vector in month t	Mkt−RF, SMB, HML
ε_{it}	Regression residual	$y_{it} - \hat{\alpha}_i - \hat{\beta}_i' f_t$
V, V_{HAC}	Covariance of the alpha vector	$N \times N$, HAC
μ, τ	Empirical-Bayes prior mean and dispersion	learned from the cross-section
L	HAC (Bartlett) lag length	$L = 12$ months
A_i^+	Frozen-beta forward alpha	next 12 months, out of sample
Q_p	Empirical p-quantile	bootstrap intervals

All vectors are column vectors unless stated otherwise. Matrix inverses in equations explain the mathematics; implementations should normally use linear solves or factorisations rather than explicitly computing an inverse.

Units and signs. Returns and factors are decimal monthly figures (a return of one percent is 0.01, not 1). A monthly decimal alpha a is reported as $a \times 12 \times 10{,}000$ annualised basis points; this is a linear reporting convention, not geometric compounding, so $a = 0.001$ becomes 120 bps/year. A *positive* alpha is benchmark-relative outperformance in-sample; a *negative* alpha is a shortfall. Ranks run from 1 (highest posterior alpha) to $N = 25$ (lowest), and quintile 5 is the highest-scoring group of five portfolios. High book-to-market is *value*; low book-to-market is *growth* — reversing these is the single most common labelling error in the course.

1.8 Recommended reading routes

1.8.1 First pass (concepts)

Chapters 3–5, 7, 10–12 and 17. Concentrate on the research question, the meaning of alpha, the joint tests, the persistence and forward chapters, the incremental decomposition, and how the paper's evidence is read. Skip the derivations on this pass and aim to state each conclusion and its main caveat in one sentence.

1.8.2 Implementation pass (code and audit)

Chapters 4–9 and 15–18. Reproduce the pipeline stage by stage — panel construction, OLS, the joint HAC covariance, the tests, empirical Bayes and the bootstrap — then the validation, the full reproducible implementation and the capstone. Audit every intermediate output against the numerical checks the chapters provide.

1.8.3 Advanced pass (theory)

Start with the mathematical foundations in Chapter 2, then the applied derivations that build on them: the joint HAC derivation (Chapter 6), the empirical-Bayes posterior and hyperparameter derivations (Chapter 8), the GRS statistic (Chapter 7), block-bootstrap theory (Chapter 9) and the complete stochastic-frontier chapter (Chapter 14). This pass fills in every "why this estimator" left implicit on the first two passes.

1.9 The governing principle

Estimate, quantify uncertainty, validate forward, then interpret. A ranking is not evidence merely because it can be computed. The research object is not the ordered list alone; it is the ordered list together with sampling uncertainty, benchmark dependence, temporal stability, and leakage-free predictive evidence.

The order of the four verbs is not cosmetic. Estimation without uncertainty invites a false hierarchy; uncertainty without forward validation cannot distinguish a persistent pattern from an in-sample artefact; and forward validation without a careful incremental decomposition cannot tell a genuinely *dynamic* forecast from a *stable* ordering that any fixed ranking would have captured. Interpretation comes last precisely because it is the step at which overclaiming is most tempting and least excusable. A recurring discipline follows from the principle: every headline number is paired with the answer to "how much could this move, and under what?".

1.10 Four layers of evidence

Layer	Question	Typical tools
Description	What happened in the historical sample?	Raw returns, factor alpha, point ranks
Inference	Could the observed pattern be sampling noise?	HAC covariance, Wald/GRS, FDR, posterior intervals
Stability	Does the conclusion survive resampling, time and model choice?	Block bootstrap, persistence, transition matrices, FF3/FF5 sensitivity
Prediction	Does information available at time t help after time t?	Frozen training estimates, forward windows, incremental tests

The layers are cumulative, not alternative. A claim earns credibility only by surviving each layer in turn: a portfolio with a large historical alpha (description) whose posterior interval excludes zero (inference), whose rank is stable under resampling and across FF3/FF5 (stability), and whose ordering carries information into strictly future data (prediction). Most of the 25 portfolios fail at the second layer; the cross-section as a whole passes the joint test but, as the incremental chapter shows, the predictive layer supports only the weaker of two possible conclusions.

1.11 Common misconceptions this course prevents

Tempting but wrong	The disciplined statement
"Alpha measures skill."	Alpha is the return the *chosen* benchmark fails to explain; on these passive test assets it is a pricing residual, not skill.
"The top-ranked portfolio is best."	Bootstrap rank intervals are wide and overlapping; the point rank is not a precise hierarchy.
"Adjacent rolling windows agree, so performance persists."	Adjacent 120-month windows overlap by 90%; the agreement is mechanical. Only non-overlapping horizons measure structural persistence.
"Past ranking predicts future returns, so the ranking has skill."	A *stable* ordering predicts too; the incremental test shows rolling re-estimation adds nothing.
"GRS rejects, so the result is solid."	GRS assumes IID-normal errors, which the diagnostics reject; the HAC/Wald test is primary here.
"Twenty-five t-tests at 5% are fine."	Twenty-five tests inflate false discoveries; the false-discovery rate must be controlled.
"The five-factor result should match the three-factor result."	Alpha is benchmark-dependent; a portfolio can move 11 ranks between FF3 and FF5.
"Our numbers match the official data."	They do not match the current official snapshot exactly; the comparison is NOT_COMPARABLE until the vintage is verified.
"The 243 bps spread is a trading profit."	It is a gross, pre-cost research diagnostic that shorts small-growth; it is not an implementable return.

1.12 Software environment and reproducibility

Every result in these notes is regenerated by a deterministic pipeline from two hashed input files. The reviewed environment is CPython 3.13 with pinned versions of NumPy, pandas, SciPy, statsmodels and matplotlib; the bootstrap uses the fixed seed 2026; and a run manifest records the input SHA-256 digests, the configuration and the runtime so that a reader can confirm exactly what produced a figure. The canonical sequence is to install the pinned requirements, run the test suite, run the analysis pipeline, export the tables, and build the report. Two properties make the pipeline auditable rather than merely runnable: the loader is *fail-closed* (it raises on any integrity violation rather than silently proceeding), and the local input files are treated as *immutable* and identified by hash, so a reference download can never quietly change the analytical sample. Chapters 15 and 16 make these mechanisms concrete.

1.13 How to use these notes

Work with a terminal and an editor open beside the text; the notes are meant to be executed, not only read. When a chapter states a number, regenerate it and confirm it matches before moving on — the habit of auditing is half of what the course teaches. Attempt each exercise in Chapter 19 before reading its guided solution, and treat the capstone in Chapter 18 as the real assessment: a complete, leakage-free reproduction of the study with a written conclusion that neither overstates nor undersells the evidence. Keep two questions in view throughout: *how uncertain is this?* and *would this claim survive an adversarial reader?* If a sentence cannot answer both, it is not yet a result.

Chapter 2

Preliminaries: mathematical and statistical foundations

This chapter assembles, with derivations, every mathematical and statistical tool that the later chapters use to answer the research question of Chapter 3. It was built the way the instructions of a good course demand: by first reading that question and its complete solution, then listing the technical machinery invoked, and finally deriving each item from first principles. Section 2.1 gives that list and maps each topic to the chapter that applies it; the remaining sections are a self-contained reference. A reader who is fluent in graduate econometrics may skim and return here whenever a later chapter cites a result.

Throughout, vectors are columns, a prime or $\top$ denotes transpose, $\mathbb{E}$, Var and Cov are the expectation, variance and covariance operators, ϕ and Φ are the standard normal density and cumulative distribution function, and I_p is the $p \times p$ identity.

2.1 The technical toolkit: what is needed and where

The research question has three parts — is there benchmark-relative performance, how uncertain is any ranking of it, and does a past ranking predict future benchmark-adjusted returns — and its answer chains together the following tools. Each row is derived in the section shown and applied in the chapter shown.

Topic	Derived in	Applied in
Quadratic forms, positive semidefiniteness, spectral theorem	§2.2	Joint HAC (Ch. 6), shrinkage (Ch. 8), conditioning
Matrix inversion lemma and partitioned inverse	§2.2	Empirical-Bayes posterior (Ch. 8)
Multivariate normal; marginal and conditional laws	§2.3	Empirical Bayes (Ch. 8), GRS (Ch. 7)
Quadratic forms of normals; χ^2 and F	§2.3	Wald and GRS tests (Ch. 7)
Ordinary least squares; projection; intercept influence	§2.4	Factor alpha (Ch. 5), HAC (Ch. 6)
Asymptotics: LLN, CLT, delta method	§2.5	All inference
Long-run variance; Newey–West HAC	§2.6	Joint alpha covariance (Ch. 6)
Wald, GRS, likelihood ratio, boundary mixtures	§2.7	Joint tests (Ch. 7), SFA (Ch. 14)
FWER, FDR, Benjamini–Hochberg	§2.8	Multiplicity control (Ch. 7)

Topic	Derived in	Applied in
Bayes, normal–normal, empirical Bayes, profile MLE	§2.9	Shrinkage (Ch. 8)
Bootstrap; block and circular block bootstrap	§2.10	Rank uncertainty (Ch. 9)
Spearman correlation; Fisher z	§2.11	Forward evaluation (Ch. 11)
Rolling vs. expanding; Diebold–Mariano; Giacomini–White	§2.12	Forward and incremental tests (Ch. 11–12)
Composed-error density; conditional mean; boundary LR	§2.13	Stochastic frontier (Ch. 14)
Excess returns and factor models	§2.14	Data and alpha (Ch. 4–5)
Numerical linear algebra and optimisation	§2.15	Implementation (Ch. 16)

2.2 Linear algebra

2.2.1 Quadratic forms and positive semidefiniteness

For a symmetric matrix $A \in \mathbb{R}^{p\times p}$ and a vector $x \in \mathbb{R}^p$, the scalar $Q(x) = x^\top Ax = \sum_{i,j} A_{ij}x_ix_j$ is a *quadratic form*. A is *positive semidefinite* (PSD), written $A \succeq 0$, if $x^\top Ax \geq 0$ for all x, and *positive definite* (PD, $A \succ 0$) if the inequality is strict for $x \neq 0$. Every covariance matrix is PSD, because for $\Sigma = \mathrm{Var}(X)$ and any fixed a,

$$a^\top \Sigma a = \mathrm{Var}(a^\top X) \geq 0. \tag{2.1}$$

This single fact recurs constantly: it is why a HAC covariance must be PSD to be trusted (Ch. 6), and why the variance of a portfolio combination of correlated alphas can be smaller than the sum of the individual variances when the covariance is negative.

2.2.2 The spectral theorem, condition number and flooring

Any symmetric A admits the *eigendecomposition*

$$A = Q\Lambda Q^\top, \qquad Q^\top Q = I_p, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p), \tag{2.2}$$

with orthonormal eigenvectors (columns of Q) and real eigenvalues λ_i. Then $x^\top Ax = \sum_i \lambda_i(q_i^\top x)^2$, so $A \succeq 0$ iff every $\lambda_i \geq 0$. Functions of A act on the eigenvalues: $A^{-1} = Q\Lambda^{-1}Q^\top$ and $A^{1/2} = Q\Lambda^{1/2}Q^\top$ when the λ_i are non-negative. The *condition number* $\kappa(A) = \lambda_{\max}/\lambda_{\min}$ measures how much relative error a linear solve can amplify; the joint HAC matrix of Chapter 6 has $\kappa \approx 139$, finite and usable. *Eigenvalue flooring* replaces each λ_i by $\max(\lambda_i, \varepsilon)$ for a small scale-relative ε, which projects a numerically-indefinite symmetric matrix onto the nearest PSD matrix in the spectral norm; the implementation floors only negligible negatives and reports how many, so a materially indefinite estimate raises rather than being silently repaired.

2.2.3 The matrix inversion lemma and the partitioned inverse

Two identities do the heavy lifting in the empirical-Bayes derivation. The *Sherman–Morrison–Woodbury* identity is

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}, \tag{2.3}$$

valid whenever the inverses exist. It is proved by multiplying the right-hand side by $A + UCV$ and simplifying to I. A consequence used repeatedly is that, for $V \succ 0$ and $\tau^2 > 0$,

$$(V^{-1} + \tau^{-2}I)^{-1} = \tau^2 I - \tau^4(V + \tau^2 I)^{-1}, \tag{2.4}$$

which is (2.3) with $A = \tau^{-2}I$, $U = V = I$, $C = V^{-1}$ after rearrangement; it lets the posterior covariance be written either as a precision inverse or in the numerically convenient "$V + \tau^2 I$" form. For a symmetric block matrix, the *partitioned inverse* uses the Schur complement $S = A_{22} - A_{21}A_{11}^{-1}A_{12}$:

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{pmatrix}. \tag{2.5}$$

This is the algebra behind the conditional normal in §2.3.

2.3 Random vectors and the multivariate normal

2.3.1 Expectation, covariance and affine maps

For a random vector X, $\mathbb{E}(X)$ is taken componentwise and $\mathrm{Cov}(X) = \mathbb{E}[(X - \mathbb{E}X)(X - \mathbb{E}X)^\top]$. The operators are linear under affine maps: if $Y = a + BX$ then

$$\mathbb{E}(Y) = a + B\,\mathbb{E}(X), \qquad \mathrm{Cov}(Y) = B\,\mathrm{Cov}(X)B^\top. \tag{2.6}$$

Both identities follow directly from linearity of expectation; the second is the reason a portfolio weight vector applied to correlated returns produces $w^\top \Sigma w$, and the reason the intercept-influence weights of §2.4 turn a residual covariance into an alpha covariance.

2.3.2 The multivariate normal density

$X \sim N(\mu, \Sigma)$ with $\Sigma \succ 0$ has density

$$f(x) = (2\pi)^{-p/2}|\Sigma|^{-1/2}\exp\Big(-\tfrac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\Big). \tag{2.7}$$

Affine images of normals are normal: by (2.6), $a + BX \sim N(a + B\mu,\, B\Sigma B^\top)$. In particular $\Sigma^{-1/2}(X - \mu) \sim N(0, I_p)$, the *whitening* transform used below.

2.3.3 Marginal and conditional normals

Partition $X = (X_1, X_2)$ with mean (μ_1, μ_2) and

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}. \tag{2.8}$$

Marginals are read off directly: $X_1 \sim N(\mu_1, \Sigma_{11})$. The conditional law is

$$X_1 \mid X_2 = x_2 \;\sim\; N\Big(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2),\; \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Big). \tag{2.9}$$

Derivation. Write the joint density and collect the terms in x_1 in the exponent. Using the partitioned inverse, the precision block acting on x_1 is $(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$, and completing the square in x_1 shows the conditional exponent is a quadratic in x_1 centred at $\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$ with that precision; the remaining terms depend only on x_2 and form the marginal, confirming the normalisation. The conditional mean is the linear regression of X_1 on X_2 and the conditional covariance is the Schur complement. This is exactly the shape of the Bayesian posterior in §2.9 and of the truncated-normal update in the frontier model of §2.13.

2.3.4 Quadratic forms of normals; χ^2 and F

If $z \sim N(0, I_p)$ then $z^\top z = \sum_i z_i^2 \sim \chi_p^2$ by definition of the chi-square law. Hence for $X \sim N(0, \Sigma)$ with $\Sigma \succ 0$, whitening gives

$$X^\top \Sigma^{-1} X = (\Sigma^{-1/2} X)^\top (\Sigma^{-1/2} X) \sim \chi_p^2. \tag{2.10}$$

This is the engine of the Wald test (§2.7): a standardised estimate that is asymptotically $N(0, \Sigma)$ under a null produces a chi-square quadratic form. If $U \sim \chi_{d_1}^2$ and $V \sim \chi_{d_2}^2$ are independent, then $F = (U/d_1)/(V/d_2) \sim F(d_1, d_2)$; the GRS statistic is exactly such a ratio because it divides a quadratic form in the alphas by an independent quadratic form in the factor means.

2.4 The linear regression model and ordinary least squares

2.4.1 Model and least-squares objective

Stack T observations of a portfolio's excess return in $y \in \mathbb{R}^T$ and place a constant and the K factors in the design $X \in \mathbb{R}^{T \times (K+1)}$, so that with $\theta = (\alpha, \beta^\top)^\top$,

$$y = X\theta + \varepsilon. \tag{2.11}$$

Ordinary least squares chooses θ to minimise the residual sum of squares $S(\theta) = (y - X\theta)^\top (y - X\theta)$.

2.4.2 Normal equations and the estimator

Expanding, $S(\theta) = y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X \theta$. Differentiating and setting the gradient to zero,

$$\frac{\partial S}{\partial \theta} = -2X^\top y + 2X^\top X\theta = 0 \implies X^\top X \hat{\theta} = X^\top y \implies \hat{\theta} = (X^\top X)^{-1} X^\top y, \tag{2.12}$$

the last step assuming X has full column rank. The Hessian $\partial^2 S / \partial\theta \partial\theta^\top = 2X^\top X \succeq 0$, so the stationary point is the global minimum.

2.4.3 Projection, residuals and the annihilator

The fitted values are $\hat{y} = X\hat{\theta} = Hy$ with the *hat matrix* $H = X(X^\top X)^{-1} X^\top$, which is symmetric and idempotent ($H^2 = H$) — the orthogonal projection onto the column space of X. The residuals are $e = (I - H)y = (I - H)\varepsilon$, and $I - H$ is the complementary projection, so residuals are orthogonal to the regressors, $X^\top e = 0$.

2.4.4 The intercept as a linear combination: influence weights

Let $c = (1, 0, \dots, 0)^\top$ select the intercept. Then $\hat{\alpha} = c^\top \hat{\theta} = c^\top (X^\top X)^{-1} X^\top y$ is a fixed linear combination of the returns:

$$\hat{\alpha} = \sum_{t=1}^{T} a_t \, y_t, \qquad a_t = c^\top (X^\top X)^{-1} x_t, \tag{2.13}$$

where $x_t^\top$ is row t of X. Because the model gives $y_t = \alpha + \beta^\top f_t + \varepsilon_t$ and the weights satisfy $\sum_t a_t = 1$, $\sum_t a_t f_t = 0$ (the intercept is orthogonal to the factors by construction), subtracting the truth yields the crucial identity

$$\hat{\alpha} - \alpha = \sum_{t=1}^{T} a_t \, \varepsilon_t. \tag{2.14}$$

The estimation error in alpha is a weighted sum of residual shocks, with weights that depend only on the common factor design. Because every one of the 25 portfolios shares the same dates and factors, the same weights a_t apply to all of them, so the same-month residual vector converts into a joint alpha error — the observation that produces the full cross-portfolio covariance in §2.6 and Chapter 6.

2.4.5 Mean, variance and the sandwich form

$\hat{\theta}$ is unbiased, $\mathbb{E}(\hat{\theta}) = \theta$, and by (2.6)

$$\mathrm{Var}(\hat{\theta}) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}, \qquad \Omega = \mathrm{Var}(\varepsilon \mid X). \tag{2.15}$$

Under the classical assumption $\Omega = \sigma^2 I$ this collapses to $\sigma^2 (X^\top X)^{-1}$, and the Gauss–Markov theorem states that OLS is then the best (minimum-variance) linear unbiased estimator. When Ω is unknown, heteroskedastic or autocorrelated — as monthly return residuals are (Ch. 13) — the "bread" $(X^\top X)^{-1}$ is kept but the "meat" $X^\top \Omega X$ is estimated robustly, giving the HAC estimator of §2.6.

2.5 Asymptotic inference

2.5.1 Laws of large numbers and the central limit theorem

For an ergodic stationary sequence with finite mean, the sample average converges in probability to the mean (LLN): $\bar{u}_T \xrightarrow{p} \mathbb{E}(u)$. If in addition the long-run variance $\Omega_\infty = \sum_{j=-\infty}^{\infty} \mathrm{Cov}(u_t, u_{t-j})$ is finite, a central limit theorem holds,

$$\sqrt{T}\,(\bar{u}_T - \mathbb{E}u) \xrightarrow{d} N(0, \Omega_\infty), \tag{2.16}$$

where the variance is the *long-run* variance, not the one-period variance, precisely because the terms are dependent. This is why forward evaluation and the Diebold–Mariano test use HAC standard errors on serially correlated window statistics (§2.12).

2.5.2 Consistency and asymptotic normality of OLS

Writing $\hat{\theta} - \theta = (X^\top X)^{-1} X^\top \varepsilon = (X^\top X/T)^{-1}(X^\top \varepsilon/T)$ and letting $Q = \operatorname{plim} X^\top X/T$, the LLN gives $X^\top \varepsilon/T \xrightarrow{p} 0$ so $\hat{\theta} \xrightarrow{p} \theta$, and the CLT applied to the score $X^\top \varepsilon/\sqrt{T}$ gives

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0,\ Q^{-1}\, S\, Q^{-1}), \qquad S = \text{long-run variance of } x_t \varepsilon_t. \tag{2.17}$$

The sandwich (2.15) is the finite-sample analogue; S is what Newey–West estimates.

2.5.3 Slutsky and the delta method

Slutsky's theorem lets consistent estimates of nuisance quantities be substituted without changing limiting distributions, which is what justifies plugging an estimated covariance into a Wald statistic. The *delta method* propagates asymptotic normality through a smooth function g: if $\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma)$ then

$$\sqrt{T}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0,\ \nabla g(\theta)^\top \Sigma\, \nabla g(\theta)). \tag{2.18}$$

The Fisher z-transform of a correlation (§2.11) is a direct application.

2.6 Dependence-robust covariance estimation

2.6.1 Autocovariance and the long-run variance

Let g_t be a mean-zero stationary (vector) score with autocovariances $\Gamma_j = \mathbb{E}[g_t g_{t-j}^\top]$. The variance of the scaled sum $T^{-1/2} \sum_{t=1}^{T} g_t$ tends to the long-run variance

$$\Omega_\infty = \sum_{j=-\infty}^{\infty} \Gamma_j = \Gamma_0 + \sum_{j=1}^{\infty} (\Gamma_j + \Gamma_j^\top), \tag{2.19}$$

which is 2π times the spectral density of g_t at frequency zero. Ignoring the cross terms (using only Γ_0) is the mistake that plain OLS standard errors make; when residuals are serially correlated the Γ_j for $j \geq 1$ are non-zero and must be included.

2.6.2 The Newey–West / Bartlett estimator

A truncated, kernel-weighted sample analogue of (2.19) is the Newey–West estimator with Bartlett weights,

$$\hat{\Omega} = \hat{\Gamma}_0 + \sum_{l=1}^{L} w_l (\hat{\Gamma}_l + \hat{\Gamma}_l^\top), \qquad w_l = 1 - \frac{l}{L+1}, \qquad \hat{\Gamma}_l = \frac{1}{T} \sum_{t=l+1}^{T} g_t g_{t-l}^\top. \tag{2.20}$$

The linearly-declining Bartlett weights are not cosmetic: they make $\hat{\Omega}$ positive semidefinite for any data. To see why, the weighted sum equals $T^{-1} \sum_{|l| \leq L} (1 - |l|/(L+1)) \sum_t g_t g_{t-l}^\top$, which is a Fejér-kernel (triangular) average of the periodogram; the Fejér kernel is non-negative, so the resulting matrix is PSD. A sharp rectangular truncation lacks this property and can yield an indefinite estimate. The bandwidth L trades bias against variance; a common data-driven choice is the automatic rule $L = \lfloor 4(T/100)^{2/9} \rfloor$, used for the forward statistics, while the monthly factor regressions use a fixed $L = 12$.

2.6.3 From single-series HAC to the joint alpha covariance

Combine (2.14) across the $N = 25$ portfolios. Let e_t be the same-month N-vector of residuals and let a_t be the common scalar intercept weight from (2.13). Define the alpha score vector $g_t = a_t e_t$. Then the stacked alpha error is $\hat{\alpha} - \alpha = \sum_t g_t$, and

$$\mathrm{Var}(\hat{\alpha} - \alpha) = \sum_t \sum_s \mathbb{E}[g_t g_s^\top] = \Gamma_0 + \sum_{l \geq 1} (\Gamma_l + \Gamma_l^\top), \tag{2.21}$$

estimated by (2.20) with the finite-sample correction $T/(T-P)$, $P = K + 1$:

$$V_{\mathrm{HAC}} = \frac{T}{T-P} \Big\{ \hat{\Gamma}_0 + \sum_{l=1}^{L} w_l (\hat{\Gamma}_l + \hat{\Gamma}_l^\top) \Big\}, \qquad \hat{\Gamma}_l = \sum_{t=l+1}^{T} g_t g_{t-l}^\top. \tag{2.22}$$

The off-diagonal entries retain contemporaneous and lagged cross-portfolio dependence; the diagonal reproduces the univariate HAC variance of each alpha, which is exactly the check performed against `statsmodels` in Chapter 6.

2.7 Hypothesis testing

2.7.1 Anatomy of a test

A test fixes a null hypothesis H_0, a statistic whose distribution is known (at least asymptotically) under H_0, and a rejection region. The *size* is the probability of rejecting a true null; the *power* is the probability of rejecting a false one. A p-value is the probability, under H_0, of a statistic at least as extreme as the one observed; it is *not* the probability that H_0 is true, and a small p-value is evidence against H_0 only within the stated model.

2.7.2 The Wald test

To test $H_0 : R\theta = r$ for a $q \times (K{+}1)$ matrix R, form

$$W = (R\hat{\theta} - r)^\top \big[R\hat{V}R^\top\big]^{-1}(R\hat{\theta} - r). \tag{2.23}$$

Under H_0, $R\hat{\theta} - r$ is asymptotically $N(0, RVR^\top)$, so by the whitening argument (2.10), $W \xrightarrow{d} \chi^2_q$. The joint test that all 25 alphas are zero is the special case $R\hat{\theta} = \hat{\alpha}$, $r = 0$:

$$W = \hat{\alpha}^\top V_{\text{HAC}}^{-1}\hat{\alpha} \;\xrightarrow{d}\; \chi^2_{25}. \tag{2.24}$$

It is the primary joint test precisely because V_{HAC} is dependence-robust: no IID or Gaussian assumption is required for the limiting chi-square.

2.7.3 The F distribution and the GRS test

Under the stronger classical assumptions — IID Gaussian residuals with a constant residual covariance Σ_e and factor covariance Σ_f with mean $\bar{f}$ — the Gibbons–Ross–Shanken statistic has an *exact* finite-sample F law,

$$\text{GRS} = \frac{T - N - K}{N} \cdot \frac{\hat{\alpha}^\top \hat{\Sigma}_e^{-1}\hat{\alpha}}{1 + \bar{f}^\top \hat{\Sigma}_f^{-1}\bar{f}} \;\sim\; F(N,\; T - N - K) \quad \text{under } H_0. \tag{2.25}$$

It is a ratio of a quadratic form in the alphas to an independent quadratic form in the factor means, i.e. a scaled Hotelling T^2, which is why it is F-distributed. Because the diagnostics of Chapter 13 reject IID normality, GRS is reported only as a conventional benchmark and the HAC/Wald test is primary.

2.7.4 Likelihood ratio, Wilks and boundary mixtures

For nested models with an unrestricted maximum log-likelihood $\hat{\ell}_1$ and a restricted one $\hat{\ell}_0$, the likelihood-ratio statistic is $\text{LR} = 2(\hat{\ell}_1 - \hat{\ell}_0)$. Wilks' theorem gives $\text{LR} \xrightarrow{d} \chi^2_q$ under H_0 when the true parameter is interior to the parameter space. When a tested parameter lies on the *boundary* — as in the frontier test of $H_0 : \sigma_u = 0$ against $\sigma_u \geq 0$ (§2.13) — the standard theory fails, because the estimator cannot move symmetrically around the null. Under regularity (Chernoff), the limiting law is the 50:50 mixture

$$\text{LR} \;\xrightarrow{d}\; \tfrac{1}{2}\chi^2_0 + \tfrac{1}{2}\chi^2_1, \qquad p = \tfrac{1}{2}\Pr(\chi^2_1 \geq \text{LR}) \text{ for } \text{LR} > 0, \tag{2.26}$$

because with probability one-half the unconstrained estimate of the non-negative parameter would have been negative and is pinned at zero (the point mass χ^2_0), and otherwise it behaves as an ordinary one-degree-of-freedom test.

2.8 Multiple testing and false-discovery control

2.8.1 Error rates

Testing m hypotheses inflates the chance of at least one false rejection. The *family-wise error rate* $\text{FWER} = \Pr(V \geq 1)$, where V is the number of true nulls rejected, is controlled conservatively by Bonferroni ($p_i \leq \alpha/m$). The *false-discovery rate* is the expected proportion of false rejections among all rejections,

$$\text{FDR} = \mathbb{E}\left[\frac{V}{\max(R, 1)}\right], \tag{2.27}$$

R being the total number of rejections. FDR is the appropriate target when many tests are screened and a few false positives are tolerable, as with 25 portfolio alphas.

2.8.2 The Benjamini–Hochberg procedure

Order the p-values $p_{(1)} \le \cdots \le p_{(m)}$. Find the largest k with

$$p_{(k)} \le \frac{k}{m}\, q, \tag{2.28}$$

and reject the hypotheses with the k smallest p-values. Under independence or positive dependence (PRDS) this controls the FDR at level $(m_0/m)q \le q$, where m_0 is the number of true nulls. Equivalently one reports monotone adjusted *q-values*

$$q_{(i)} = \min_{k \ge i}\ \min\Big(1,\ \frac{m\, p_{(k)}}{k}\Big), \tag{2.29}$$

a reverse cumulative minimum that guarantees $q_{(1)} \le \cdots \le q_{(m)}$; rejecting when $q_i \le q$ reproduces the step-up rule. The study applies (2.29) in two separate families — raw-alpha p-values and the 25 frontier boundary p-values — because they test different nulls.

2.9 Bayesian shrinkage and empirical Bayes

2.9.1 Bayes' theorem and conjugacy

Bayes' theorem combines a likelihood and a prior into a posterior, $p(\theta \mid x) \propto p(x \mid \theta)\, p(\theta)$. A prior is *conjugate* when the posterior stays in the same family; the normal likelihood with a normal prior is the canonical case and yields closed-form shrinkage.

2.9.2 The normal–normal posterior by completing the square

Scalar case. Let $x \mid \theta \sim N(\theta, v)$ and $\theta \sim N(\mu, \tau^2)$. The log-posterior in θ is, up to constants,

$$-\tfrac{1}{2}\Big[\frac{(x-\theta)^2}{v} + \frac{(\theta-\mu)^2}{\tau^2}\Big] = -\tfrac{1}{2}\Big[\theta^2\Big(\tfrac{1}{v} + \tfrac{1}{\tau^2}\Big) - 2\theta\Big(\tfrac{x}{v} + \tfrac{\mu}{\tau^2}\Big)\Big] + \text{const.} \tag{2.30}$$

Completing the square shows $\theta \mid x \sim N(m, s^2)$ with precision $s^{-2} = v^{-1} + \tau^{-2}$ and

$$m = \frac{v^{-1}x + \tau^{-2}\mu}{v^{-1} + \tau^{-2}} = \mu + \frac{\tau^2}{\tau^2 + v}\,(x - \mu). \tag{2.31}$$

The posterior mean is a precision-weighted average of data and prior; the *shrinkage weight* $w = \tau^2/(\tau^2 + v)$ is near one when the datum is precise (v small) and near zero when it is noisy.

Multivariate case. With $\hat\alpha \mid \alpha \sim N(\alpha, V)$ and $\alpha \sim N(\mu\mathbf{1}, \tau^2 I)$, the same completion of the square in the vector α gives posterior precision $P = V^{-1} + \tau^{-2} I$ and

$$m = \mu\mathbf{1} + \tau^2\big(V + \tau^2 I\big)^{-1}(\hat\alpha - \mu\mathbf{1}), \qquad C = \tau^2 I - \tau^4\big(V + \tau^2 I\big)^{-1}, \tag{2.32}$$

where the second form of C follows from the Woodbury identity (2.4). Because V is not diagonal, the update is genuinely multivariate: one portfolio's posterior depends on uncertainty it shares with the others.

2.9.3 The winner's curse and James–Stein

Selecting the largest of N noisy estimates biases it upward, because extreme observed values combine signal with favourable noise. Shrinkage corrects this: the James–Stein estimator, which shrinks the sample mean vector toward a common point, has uniformly smaller risk than the raw estimator in dimension $N \ge 3$. Empirical-Bayes shrinkage of the 25 alphas is the applied form of this dominance and is why the posterior ranking is more trustworthy than the raw-alpha ranking.

2.9.4 Empirical Bayes: estimating μ and τ

The prior parameters are not known; empirical Bayes estimates them from the same cross-section by maximising the *marginal* likelihood. Integrating α out of the hierarchy gives

$$\hat{\alpha} \sim N(\mu\mathbf{1},\ \Sigma_\tau), \qquad \Sigma_\tau = V + \tau^2 I. \tag{2.33}$$

For fixed τ, the marginal log-likelihood is quadratic in μ and its first-order condition is a generalised-least-squares mean,

$$\frac{\partial \ell}{\partial \mu} = \mathbf{1}^\top \Sigma_\tau^{-1}(\hat{\alpha} - \mu\mathbf{1}) = 0 \implies \hat{\mu}(\tau) = \frac{\mathbf{1}^\top \Sigma_\tau^{-1}\hat{\alpha}}{\mathbf{1}^\top \Sigma_\tau^{-1}\mathbf{1}}. \tag{2.34}$$

Substituting $\hat{\mu}(\tau)$ back gives a one-dimensional *profile* log-likelihood in $\tau^2 \geq 0$,

$$\ell_p(\tau) = -\tfrac{1}{2}\log|\Sigma_\tau| - \tfrac{1}{2}(\hat{\alpha} - \hat{\mu}(\tau)\mathbf{1})^\top \Sigma_\tau^{-1}(\hat{\alpha} - \hat{\mu}(\tau)\mathbf{1}), \tag{2.35}$$

maximised by bounded one-dimensional optimisation (§2.15). The fitted values here are $\hat{\mu} \approx -26$ and $\hat{\tau} \approx 94$ annualised basis points. Reported posterior intervals add a first-order term for uncertainty in $\hat{\mu}$ but do not integrate over τ, so they are model-based summaries, not exact guarantees.

2.10 Resampling and the bootstrap

2.10.1 The plug-in principle

The bootstrap approximates the sampling distribution of a statistic $\hat{\vartheta} = s(\text{data})$ by resampling from an estimate of the data-generating process and recomputing s many times. For independent data one resamples observations with replacement; the empirical distribution of the recomputed statistics approximates its true sampling distribution. Crucially, the *whole* estimator is recomputed on each resample, so uncertainty in every stage is captured.

2.10.2 Block and circular block bootstraps

Monthly returns are dependent, so resampling single months would destroy serial and cross-sectional structure. The *moving block bootstrap* resamples contiguous blocks of length b; concatenating $\lceil T/b \rceil$ blocks preserves dependence within a block. The *circular* variant wraps the index modulo T,

$$\text{block starting at } s: \quad ((s + h - 1) \bmod T) + 1, \quad h = 0, \dots, b-1, \tag{2.36}$$

so that every observation — including those near the start and end of the sample — is equally likely to appear, removing the edge bias of the non-circular version. Consistency requires $b \to \infty$ with $b/T \to 0$; a 12-month block is used here. Applying the *same* resampled date index to all 25 portfolios (a *common-date* bootstrap) preserves contemporaneous cross-portfolio shocks.

2.10.3 Bootstrapping ranks and quantiles

A rank is a discontinuous function of the scores, so its uncertainty is not described by a standard error. The bootstrap sidesteps this: within each replicate r the full pipeline — regressions, joint HAC covariance, empirical-Bayes hyperparameters — is refitted and the portfolios re-ranked to give $R_i^{(r)}$. The rank interval is the pair of empirical quantiles

$$\left[\, Q_{0.025}\{R_i^{(r)}\},\ Q_{0.975}\{R_i^{(r)}\} \,\right], \tag{2.37}$$

and the top-quintile probability is the fraction of replicates with $R_i^{(r)} \leq 5$. These are resampling distributions, not Gaussian intervals, and can be asymmetric and bounded at ranks 1 and 25.

2.11 Rank statistics

2.11.1 Spearman rank correlation

Spearman's ρ is the Pearson correlation computed on the ranks of two variables; it measures monotone association and is invariant to any increasing transformation of either variable. In the forward evaluation it compares the cross-section of training scores with the cross-section of future alphas, so that only the *ordering*, not the scale, is tested.

2.11.2 The Fisher z-transformation

A sample correlation near ± 1 has a strongly skewed, bounded sampling distribution, so correlations are averaged on the *Fisher* scale

$$z = \operatorname{atanh}(\rho) = \tfrac{1}{2} \log \frac{1+\rho}{1-\rho}. \tag{2.38}$$

This is variance-stabilising. By the delta method (2.18) with $g(\rho) = \operatorname{atanh}(\rho)$, $g'(\rho) = 1/(1-\rho^2)$, and the approximate correlation variance $\operatorname{Var}(\hat{\rho}) \approx (1-\rho^2)^2/n$,

$$\operatorname{Var}(z) \approx g'(\rho)^2 \operatorname{Var}(\hat{\rho}) \approx \frac{1}{n}, \tag{2.39}$$

free of ρ to first order. Window correlations are therefore averaged as z, given HAC standard errors across windows, and mapped back with tanh.

2.12 Out-of-sample forecast evaluation

2.12.1 Rolling vs. expanding estimation and look-ahead bias

A *rolling* scheme estimates on a fixed-length trailing window $[t-w+1, t]$ and predicts the next h months; an *expanding* scheme uses all data $[1, t]$. *Look-ahead bias* is any leakage of information dated after t into the time-t prediction — a future return entering a score, rank, prior parameter or factor loading. The study freezes every such quantity at the training cutoff and evaluates only strictly later months, and asserts the non-overlap of evaluation windows so that annual statistics are not mechanically correlated through shared future data.

2.12.2 The Diebold–Mariano test

To compare two forecasts with losses $L(e_{1t})$ and $L(e_{2t})$, form the loss differential $d_t = L(e_{1t}) - L(e_{2t})$ and test $H_0 : \mathbb{E}(d_t) = 0$ (equal predictive accuracy) with

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\operatorname{Var}}_{\mathrm{HAC}}(\bar{d})}} \xrightarrow{d} N(0,1), \tag{2.40}$$

where the HAC variance (2.20) is needed because d_t is serially correlated. The incremental test of Chapter 12 is exactly this: d_t is the per-window difference between the rolling ranking's predictive score (rank correlation, or quintile spread) and a benchmark ordering's, and the reported HAC t-statistic on $\bar{d}$ is a Diebold–Mariano statistic.

2.12.3 Giacomini–White conditional predictive ability

The classical Diebold–Mariano theory treats forecasts as given. Giacomini and White extend equal-accuracy testing to forecasts produced by models that are *re-estimated* on a rolling window of fixed length, which is precisely the design here; their result justifies treating the rolling-versus-benchmark differential as a valid test of conditional predictive ability rather than merely of two fixed forecasts.

2.13 Composed-error (stochastic frontier) models

2.13.1 The composed error and its density

The half-normal stochastic-frontier model writes the residual as $\varepsilon = v - u$ with $v \sim N(0, \sigma_v^2)$ and a one-sided $u = |w|$, $w \sim N(0, \sigma_u^2)$, independent. Its density is the convolution

$$f(\varepsilon) = \int_0^\infty f_v(\varepsilon + u)\, f_u(u)\, du = \frac{2}{\sigma}\,\phi\Big(\frac{\varepsilon}{\sigma}\Big)\,\Phi\Big(-\frac{\lambda\varepsilon}{\sigma}\Big), \qquad \sigma^2 = \sigma_v^2 + \sigma_u^2,\ \lambda = \frac{\sigma_u}{\sigma_v}. \tag{2.41}$$

Derivation. The integrand is $\propto \exp\{-\frac{1}{2}[(\varepsilon+u)^2/\sigma_v^2 + u^2/\sigma_u^2]\}$; completing the square in u shows u enters as a normal kernel with mean $\mu_* = -\varepsilon\sigma_u^2/\sigma^2$ and variance $\sigma_*^2 = \sigma_v^2\sigma_u^2/\sigma^2$, and the leftover ε terms assemble into $\phi(\varepsilon/\sigma)$; integrating the normal kernel over $u \geq 0$ contributes the factor $\Phi(\mu_*/\sigma_*) = \Phi(-\lambda\varepsilon/\sigma)$, giving the stated density (the factor 2 is the half-normal normalisation).

2.13.2 The conditional mean $\mathbb{E}[u \mid \varepsilon]$

From the completion above, $u \mid \varepsilon$ is normal with mean μ_* and variance σ_*^2 *truncated* to $[0, \infty)$. The mean of a normal $N(\mu_*, \sigma_*^2)$ truncated below at zero is the Jondrow et al. formula

$$\mathbb{E}[u \mid \varepsilon] = \mu_* + \sigma_*\,\frac{\phi(\mu_*/\sigma_*)}{\Phi(\mu_*/\sigma_*)}, \tag{2.42}$$

the standard truncated-normal mean (a location plus the inverse Mills ratio scaled by σ_*). This is the portfolio-specific estimate of the one-sided shortfall used, cautiously, as a residual-asymmetry diagnostic.

2.13.3 The boundary likelihood-ratio test

Testing whether a distinct one-sided component exists is $H_0 : \sigma_u = 0$, a value on the boundary of the admissible set $\sigma_u \geq 0$. As in (2.26), the likelihood-ratio statistic against the Gaussian model has the $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ limiting law, so the p-value is $\frac{1}{2}\Pr(\chi_1^2 \geq \mathrm{LR})$. Only one of the 25 portfolios survives this test after false-discovery control, which is why the frontier apparatus remains a narrow diagnostic rather than a ranking.

2.14 Asset-pricing setup

2.14.1 What "Fama–French" means

The names attached to the model and the data are those of the financial economists Eugene F. Fama (a Nobel laureate in economics, 2013) and Kenneth R. French, whose joint work defines the benchmark used throughout these notes. Understanding the term requires one step of background.

The starting point: the CAPM. The classical Capital Asset Pricing Model (Sharpe, Lintner) holds that a single systematic factor — the excess return of the aggregate market — prices all assets: an asset's expected excess return should be proportional to its market "beta,"

$$\mathbb{E}[y_i] = \beta_i\,\mathbb{E}[\mathrm{Mkt} - \mathrm{RF}], \tag{2.43}$$

with intercept zero. Empirically this fails in a systematic way. Two groups of stocks earn persistently higher average returns than their market beta explains: *small* firms (low market capitalisation) and *value* firms (high book-to-market ratio). These are the "size" and "value" anomalies.

The three-factor model (FF3). Fama and French (1993) proposed augmenting the market factor with two *factor-mimicking portfolios* built precisely to capture those anomalies: SMB ("small minus big," long small firms and short large ones) and HML ("high minus low," long value and short growth). "FF3" abbreviates the resulting Fama–French three-factor model. It has been the standard empirical risk benchmark in finance for three decades, which is exactly why it is the benchmark here: alpha measured against FF3 is the conventional definition of return "unexplained" by known systematic exposures.

The five-factor extension (FF5). Fama and French (2015) added two further factors — RMW ("robust minus weak" profitability) and CMA ("conservative minus aggressive" investment) — to capture patterns FF3 still missed. "FF5" is that five-factor model. The study uses FF3 as its canonical benchmark and FF5 as a prespecified sensitivity check, precisely because alpha is benchmark-dependent (§3.3).

The test assets and the data source. The 25 portfolios analysed here are themselves a Fama–French construction: all NYSE/AMEX/NASDAQ stocks are sorted, at each formation date, into five size groups and five book-to-market groups, and the $5 \times 5 = 25$ intersections are held as value-weighted portfolios. Both the factor series and the 25 portfolios are published and maintained in the *Kenneth R. French Data Library*, the canonical public source from which the local inputs are drawn (Chapter 4). A subtle but important consequence follows: because SMB and HML are built *from* sorts on size and book-to-market, and the 25 test assets are built from the *same* sorts, regressing these portfolios on FF3 measures the model against its own raw material — another reason a non-zero alpha is a pricing residual, not evidence of skill.

2.14.2 Excess returns and factor models

The one-period excess return is $y_{it} = r_{it} - r_{ft}$, the return above the risk-free rate. A linear factor model writes it as $y_{it} = \alpha_i + \beta_i^\top f_t + \varepsilon_{it}$, where f_t collects traded factor returns. For the three-factor model $f_t = (\text{Mkt} - \text{RF}, \text{SMB}, \text{HML})_t^\top$: the market excess return, SMB (small minus big) and HML (high minus low book-to-market); the five-factor model appends RMW (profitability) and CMA (investment). Every component is itself an excess or long–short return, so the regression places dependent variable and regressors on a common basis.

2.14.3 Alpha as a model-relative residual

The intercept α_i is the average excess return the chosen factors leave unexplained. It is model-relative: a non-zero alpha signals mispricing *relative to the specified factors*, and can arise from an omitted priced risk, a friction, a nonlinearity, or genuine abnormal performance. On passively sorted test assets it is a pricing residual, not evidence of skill — the interpretive discipline that the whole course enforces.

2.15 Numerical methods

2.15.1 Solve, do not invert

Equations are written with matrix inverses for clarity, but a stable implementation solves linear systems instead. For a PD matrix the Cholesky factorisation $A = LL^\top$ (L lower-triangular) reduces $Ax = b$ to two triangular solves, at lower cost and better accuracy than forming A^{-1}. The log-determinant needed for the marginal likelihood (2.33) is $\log|A| = 2\sum_i \log L_{ii}$.

2.15.2 Conditioning and eigenvalue flooring

The relative error in a solve is bounded by roughly $\kappa(A)$ times the input error, so an ill-conditioned covariance loses precision. The pipeline symmetrises every estimated covariance, inspects its eigenvalues, floors only negligible negative eigenvalues by a scale-relative amount (§2.2), and raises if the matrix is materially indefinite, so numerical repair is auditable rather than silent.

2.15.3 Positivity and bounded optimisation

Scale parameters must stay positive, so they are estimated on the log scale, e.g. $\sigma = e^{\eta}$ with η unconstrained, which turns a constrained problem into an unconstrained one and keeps the optimiser inside the feasible region. The one-dimensional profile in τ^2 (2.34) is maximised by a bounded scalar routine (Brent's method) on $[0, \text{upper}]$, and the frontier likelihood uses a robust Newton/quasi-Newton step with the analytic score of Chapter 14.

Chapter 3

The research question

The study examines 25 portfolios formed from the intersections of five size groups and five book-to-market groups. It asks whether their returns contain persistent performance unexplained by a standard factor benchmark, how uncertain any resulting ranking is, and whether a ranking estimated from the past predicts genuinely future benchmark-adjusted returns. This chapter states those questions precisely, fixes what would count as an answer to each, and explains why the exercise is far harder than running twenty-five regressions and reading off the largest intercept. The tools it invokes are all derived in Chapter 2; here they are assembled into a single research design.

3.1 Formalising the estimand and the three questions

Let $i = 1, \ldots, N$ index the $N = 25$ portfolios and $t = 1, \ldots, T$ the $T = 1{,}193$ months. For each portfolio the benchmark regression of Chapter 2 is

$$y_{it} = \alpha_i + \beta_i^\top f_t + \varepsilon_{it}, \qquad y_{it} = r_{it} - r_{ft}, \tag{3.1}$$

with f_t the FF3 factor vector. The population object of interest is the N-vector of intercepts $\alpha = (\alpha_1, \ldots, \alpha_N)^\top$ and, derived from it, the ranking that orders the portfolios from highest to lowest. Everything the study reports is a functional of α and of its estimation uncertainty. The research question decomposes into three sharply different statistical problems.

	Question	Formal object	What counts as an answer
Q1	Is there benchmark-relative performance at all?	The vector α and the joint null $H_0 : \alpha = 0$	A dependence-robust rejection of H_0 and per-portfolio evidence surviving multiplicity control
Q2	How well-determined is the ranking?	The rank vector $R = \text{rank}(\hat{\alpha}^{\text{post}})$ as a random object	An honest sampling/resampling distribution of each rank, not a point list
Q3	Does a past ranking predict future benchmark-adjusted returns, and does re-estimating it help?	The forward association between a frozen ranking and later alpha, and its *increment* over a fixed ordering	A strictly leakage-free, dependence-robust test, and a decomposition separating dynamic from static content

Q1 is a question of *inference* (Chapters 6–7); Q2 is a question of *estimation uncertainty* (Chapters 8–9); Q3 is a question of *prediction* (Chapters 11–12). They are logically independent: one can reject $H_0 : \alpha = 0$ (Q1) yet find the ranking almost undetermined (Q2) and its dynamic content null (Q3), which is in fact what happens.

3.2 Why "performance" is model-relative

The estimand α_i is defined only *relative to the chosen factors*. Writing the population projection of y_i on $(1, f)$, the intercept is

$$\alpha_i = \mathbb{E}[y_{it}] - \beta_i^\top \mathbb{E}[f_t], \tag{3.2}$$

so α_i is the average excess return left after removing exposure to *those* factors and nothing more. A non-zero α_i can arise from an omitted priced risk, a friction, a nonlinear payoff, a change in factor loadings over time, or genuine abnormal performance — the regression cannot tell these apart. On the 25 test assets the interpretation is even narrower: these are the very portfolios from which the size and value factors are constructed, so their α is a *pricing residual* of the model against its own building blocks, not a measure of skill. The course therefore never writes "skill" for α; it writes "benchmark-relative" and treats benchmark choice itself as a variable (the FF3-versus-FF5 sensitivity of Chapter 13).

3.3 Why this is harder than running 25 regressions

A naive analysis would run (3.1) twenty-five times, take the largest $\hat{\alpha}_i$, and declare a winner. Seven features of the problem make that invalid; each is quantified below and handled by a specific tool from Chapter 2.

3.3.1 1. The alpha estimates are correlated

The portfolios share calendar dates, common market shocks and overlapping construction rules, so their residuals ε_{it} are correlated across i within a month. By the influence-weight identity (2.13), $\hat{\alpha}_i - \alpha_i = \sum_t a_t \varepsilon_{it}$ with weights common to all portfolios, so the alpha errors inherit that cross-sectional dependence and the estimator's covariance is a *full* matrix V, not a diagonal one. This matters concretely: the precision of a comparison between two portfolios is

$$\mathrm{Var}(\hat{\alpha}_i - \hat{\alpha}_j) = V_{ii} + V_{jj} - 2V_{ij}, \tag{3.3}$$

which positive covariance V_{ij} can make far smaller — or, if negative, larger — than the sum of the marginal variances. Twenty-five separate standard errors discard every V_{ij} and therefore misstate both joint tests and rankings. The remedy is the joint HAC covariance of §2.6 and Chapter 6.

3.3.2 2. Selecting the largest estimate is a winner's curse

Even if every true α_i were equal, the largest *observed* estimate would exceed the truth, because extreme draws combine signal with favourable noise. In the simplest case $\hat{\alpha}_i = \alpha_i + e_i$ with $e_i \sim N(0, \sigma^2)$ independent, the expected maximum of the noise alone is

$$\mathbb{E}\Big[\max_{i \le N} e_i\Big] \;\approx\; \sigma\sqrt{2 \ln N}, \tag{3.4}$$

so the top-ranked estimate is biased upward by roughly $\sigma\sqrt{2 \ln 25} \approx 2.54\,\sigma$ in this idealisation. Reporting the raw maximum therefore overstates the leader and, symmetrically, understates the laggard. The correction is partial pooling: the empirical-Bayes shrinkage of §2.9 pulls noisy extremes toward the common mean by the weight $\tau^2/(\tau^2 + v)$ of (2.31), and the James–Stein dominance result guarantees this reduces risk in dimension $N \ge 3$.

3.3.3 3. Testing twenty-five hypotheses inflates false discoveries

Screening $N = 25$ nulls at level $\alpha = 0.05$ each makes at least one false rejection likely even when all nulls hold: for independent tests

$$\Pr(\text{at least one false positive}) = 1 - (1 - 0.05)^{25} \approx 0.72, \tag{3.5}$$

and the expected number of false positives is $N\alpha = 1.25$. A "5% significant" alpha found this way is uninformative on its own. The study controls the false-discovery rate with the Benjamini–Hochberg procedure of §2.8, reporting monotone q-values (2.29), and keeps the raw-alpha family separate from the frontier-test family because they test different nulls.

3.3.4 4. Ranks are discontinuous

The rank of portfolio i,

$$R_i = 1 + \sum_{j \neq i} \mathbf{1}\{\hat{s}_j > \hat{s}_i\}, \tag{3.6}$$

is a step function of the scores $\hat{s}$: it is flat almost everywhere and jumps by an integer when two scores cross. Its derivative is zero wherever it is defined, so the delta method (2.18) cannot supply a standard error, and an arbitrarily small change in a score can move a rank by several positions. Rank uncertainty must therefore be obtained by resampling the *entire* estimator, which is what the common-date circular block bootstrap of §2.10 and Chapter 9 does; the result is a rank interval, not a point.

3.3.5 5. Overlapping windows manufacture persistence

The rolling analysis uses windows of length $w = 120$ months advanced by a step $s = 12$. Two windows k steps apart share $\max(0, w - ks)$ months, an overlap fraction

$$\frac{w - ks}{w} = 1 - \frac{ks}{w}, \qquad ks < w, \tag{3.7}$$

so *adjacent* windows ($k = 1$) share $108/120 = 90\%$ of their data. A high correlation between adjacent rolling estimates is then largely mechanical: the two estimates are computed from almost the same months. Structural persistence can only be read from *non-overlapping* comparisons, which require $ks \geq w$, i.e. $k \geq w/s = 10$ steps $= 120$ months apart. Chapter 10 gives structural weight only to those disjoint comparisons, where persistence turns out to be weak.

3.3.6 6. A forward relation need not be dynamic skill

This is the subtle point on which the study turns. Suppose each cell has a *fixed* benchmark-relative level c_i (small-growth structurally poor, small-value structurally strong) so that both the training score and the future alpha are noisy readings of the same c_i:

$$\hat{s}_{it} = c_i + u_{it}, \qquad A^{+}_{i,t} = c_i + w_{i,t}. \tag{3.8}$$

Then the cross-sectional correlation $\mathrm{Corr}_i(\hat{s}_{it}, A^{+}_{i,t})$ is positive purely through the shared fixed effect c_i, with *no time-varying information* in the ranking at all. A positive forward correlation is thus consistent with a merely *stable* ordering. Distinguishing this from genuine dynamic forecasting requires comparing the time-varying rolling ranking against a fixed ordering on the same future data — the incremental test of §2.12 and Chapter 12, which is a Diebold–Mariano test of equal predictive accuracy (2.40).

3.3.7 7. Alpha depends on the benchmark

Because $\alpha_i = \mathbb{E}[y_i] - \beta_i^\top \mathbb{E}[f]$ depends on which factors f are included, enlarging the model changes both the loadings and the spanned space and hence the intercept. A portfolio that loads on a priced dimension absent from FF3 will show a spurious FF3 alpha that shrinks or reverses under FF5. The study measures this directly: on a common sample the FF3 and FF5 rankings correlate but a single portfolio moves eleven rank positions (Chapter 13). Benchmark dependence is a property of the estimand, not a nuisance to be tuned away.

3.4 The central logic: stable ordering versus dynamic skill

The seven difficulties combine into one argument, which the reader should be able to reconstruct and, above all, to stop at the correct conclusion.

1. Historical benchmark-relative alpha varies across the cross-section (description, Chapter 5).
2. A ranking formed from past alpha correlates with future benchmark-adjusted returns (forward validation, Chapter 11).
3. But by difficulty 6 this is also what a *fixed* ordering would produce, so the rolling ranking is compared against static and expanding-window orderings on identical future data (incremental test, Chapter 12).
4. The rolling ranking adds no measurable increment over the fixed ordering: the Diebold–Mariano differential is statistically indistinguishable from zero.
5. Conclusion: the predictable component is a *stable* cross-sectional ordering of persistent characteristic mispricing, *not* dynamic timing skill.

Formally, the study pits two hypotheses against each other. Let ρ_t^{roll} and ρ_t^{fix} be the window-t predictive scores of the rolling and fixed orderings and $d_t = \rho_t^{\text{roll}} - \rho_t^{\text{fix}}$.

$$H_{\text{stable}}: \ \mathbb{E}(d_t) \leq 0 \qquad \text{versus} \qquad H_{\text{dynamic}}: \ \mathbb{E}(d_t) > 0. \tag{3.9}$$

The forward test of Q3 establishes only that $\mathbb{E}(\rho_t^{\text{roll}}) > 0$; it is the incremental test that adjudicates H_{stable} against H_{dynamic}, and it fails to reject H_{stable}. A student must be able to distinguish the weak, supported claim "a past ranking correlates with future alpha" from the strong, unsupported claim "re-estimating a rolling ranking creates useful dynamic forecasts." Conflating the two is the central error the whole design is built to prevent.

3.5 What would count as evidence, and what the study does not claim

Each question is tied to a layer of evidence from Chapter 1, and each has a clear failure mode. Q1 is answered only if a *dependence-robust* joint test rejects $H_0 : \alpha = 0$ and individual alphas survive false-discovery control; a rejection under an IID-Gaussian test alone (GRS) would not suffice, because the diagnostics reject those assumptions (Chapter 13). Q2 is answered by a rank *distribution*; a point ranking with no interval is not an answer. Q3 is answered only by a strictly leakage-free forward test *and* an incremental decomposition; in-sample fit, or a forward correlation without the incremental comparison, would leave H_{stable} and H_{dynamic} unseparated.

Finally, the study is explicit about its non-goals. It does not claim an implementable trading profit — the reported spreads are gross of costs and short hard-to-borrow small-growth stocks; it does not claim causal skill; and it does not claim that its local data exactly match the current official source, a discrepancy left visible as NOT_COMPARABLE (Chapter 4). Stating what is *not* concluded is part of answering the research question honestly.

Chapter 4

Data, portfolio construction, units and provenance

This chapter fixes the raw material of the whole study: the analytical grain of a single observation, the units in which returns and factors are measured, how the 25 test portfolios and the factors are constructed, the fail-closed checks that a panel must pass before any regression is run, and why reconciling a file against its official source is a different question from reproducing a number from that file. The estimator of Chapter 2 is only as trustworthy as the panel it consumes, so the discipline here is not clerical — it is part of the statistical specification.

4.1 The 5 × 5 test-asset grid

At each annual formation date, every eligible US common stock (NYSE, AMEX and NASDAQ) is placed into one of five *size* groups by market equity and, independently, one of five *book-to-market* groups. The $5 \times 5 = 25$ intersections define 25 portfolios whose monthly returns are *value-weighted*: each constituent enters in proportion to its market equity, so a handful of large firms can dominate a cell. The breakpoints are recomputed each year from NYSE stocks only, and a stock stays in its assigned cell until the next formation date. A label such as `SMALL HiBM` therefore denotes a *characteristic cell* — small firms with a high book-to-market ratio — not a company, a managed fund, or a trading recommendation. The corner cells name the extremes: `SMALL LoBM` is small-growth, `SMALL HiBM` is small-value, `BIG LoBM` is large-growth and `BIG HiBM` is large-value; the interior cells are written `ME`j `BM`k for size quintile j and book-to-market quintile k.

Two properties of this grid drive design choices later. First, the cells are *not independent entities*: neighbours share construction rules and, through the common market, share monthly shocks — the source of the cross-sectional dependence handled by the joint covariance of Chapter 2 (§2.6). Second, the grid is built from the very sorts (size and book-to-market) that define the SMB and HML factors, so regressing these portfolios on FF3 tests the model against its own raw material (§2.14.1).

Term	Meaning	Common mistake
Size	Market equity grouping	Confusing "small" with low price rather than low market capitalisation
Book-to-market	Book equity divided by market equity	Reversing value and growth: high B/M is value; low B/M is growth

Term	Meaning	Common mistake
Value-weighted return	Constituent returns weighted by market value	Accidentally parsing the equal-weighted block
Portfolio-month	One portfolio observed in one calendar month	Joining files many-to-many and duplicating this grain

4.2 Returns and factors

The dependent variable of every regression is the portfolio's monthly *excess* return,

$$y_{it} = r_{it} - r_{ft}, \tag{4.1}$$

the total return r_{it} of portfolio i in month t minus the one-month risk-free rate r_{ft}. The regressors are the traded factor returns collected in f_t:

- **Mkt-RF:** the aggregate market return in excess of the risk-free rate (itself an excess return).
- **SMB** (small minus big): the size factor, long small firms and short large ones.
- **HML** (high minus low): the value factor, long high book-to-market and short low.
- **RMW and CMA:** profitability (robust minus weak) and investment (conservative minus aggressive) factors, added by the five-factor model.

Because each regressor is itself an excess or self-financing long–short return, the dependent variable and the regressors sit on the same economic footing, and the intercept α_i inherits the units of a monthly excess return (§2.14).

Units and annualisation. The canonical analysis works in *decimal monthly* returns, so a one-percent monthly return is 0.01, not 1. A monthly decimal alpha a is reported in annualised basis points as $a \times 12 \times 10{,}000$; for example $a = 0.001$ becomes $0.001 \times 12 \times 10{,}000 = 120$ bps/year. This is a *linear* reporting convention, not geometric compounding, chosen so that the reported figure is a transparent scaling of the estimated coefficient. Mixing percent and decimal units is the most common unit error and would inflate every alpha and standard error by a factor of one hundred; the loader therefore records and checks the units explicitly.

4.3 Finance foundations for a mathematics graduate

4.3.1 Simple and portfolio returns

If an asset begins the month at price $\mathrm{P_{t-1}}$, pays dividend $\mathrm{D_t}$, and ends at $\mathrm{P_t}$, its simple total return is

$$R_t = (P_t + D_t - P_{t-1})/P_{t-1}. \tag{4.2}$$

For J constituents with beginning-of-period portfolio weights $\mathrm{w_{jt-1}}$, $\sum_{\mathrm{j}}\mathrm{w_{jt-1}}$=1, the one-period portfolio return is

$$R_{\mathrm{pt}} = \sum_{j=1}^{J} w_{jt-1} R_{\mathrm{jt}}. \tag{4.3}$$

In a value-weighted portfolio, $\mathrm{w_{jt-1}}$ is proportional to the firm's market equity at the weighting date. Larger companies therefore receive larger weights.

4.3.2 Why subtract the risk-free rate?

A risky investment must first compensate the investor for deferring consumption. The risk-free return r_{ft} is the baseline return available without bearing the portfolio's risky exposure. The excess return $y_{it}=r_{it}-r_{ft}$ is the part to be explained by risky factors and alpha.

4.3.3 Market equity and book-to-market

$$\text{Market equity} = \text{share price} \times \text{shares outstanding}, \tag{4.4}$$

$$\text{Book-to-market} = \frac{\text{accounting book equity}}{\text{market equity}}. \tag{4.5}$$

Market equity ("size") measures how large the firm is in the market; book-to-market compares the accountant's valuation of the firm's equity to the market's. High book-to-market companies are conventionally called *value* stocks — the market prices them low relative to book — and low book-to-market companies are called *growth* stocks. These labels describe *characteristics*, not guaranteed future returns, and the value/growth direction is the single most common labelling error: high B/M is value, low B/M is growth.

4.3.4 How the FF3 factor portfolios are constructed

The 25 portfolios in this study are 5×5 **test assets**. They are related to, but not identical with, the six 2×3 portfolios used in the standard construction of the SMB and HML *factors*. Writing S/B for the two size groups (small, big) and $L/N/H$ for the three book-to-market groups (low, neutral, high), the six building-block portfolios are the intersections, and the textbook factor formulas average them:

$$\text{SMB} = \tfrac{1}{3}(S/L + S/N + S/H) - \tfrac{1}{3}(B/L + B/N + B/H), \tag{4.6}$$

$$\text{HML} = \tfrac{1}{2}(S/H + B/H) - \tfrac{1}{2}(S/L + B/L). \tag{4.7}$$

SMB is thus the average small-cap return minus the average large-cap return holding book-to-market spread fixed, and HML is the average high-B/M return minus the average low-B/M return holding size fixed. Both are *self-financing long–short* returns — the long and short legs carry offsetting capital weights — so they are returns on zero-cost portfolios, and Mkt-RF is likewise an excess return. The five-factor model appends two more long–short returns built the same way,

$$\text{RMW} = (\text{robust-profitability}) - (\text{weak-profitability}), \tag{4.8}$$

$$\text{CMA} = (\text{conservative-investment}) - (\text{aggressive-investment}). \tag{4.9}$$

The distinction between the 2×3 factor-building portfolios and the 5×5 test assets matters: the study regresses the *test assets* on the *factors*, and because both come from size and book-to-market sorts the resulting alpha is a residual of the model against its own inputs.

Economic meaning of the regression. The regression $y_{it} = \alpha_i + \beta_i^\top f_t + \varepsilon_{it}$ asks how much of a portfolio's excess return moves one-for-one with the broad market, size and value returns; the loadings β_i are those sensitivities and the intercept α_i is the average return left after the fitted linear co-movement is removed. A positive $\beta_{i,\mathrm{HML}}$, for instance, says the cell behaves like a value portfolio, which the sanity checks of Chapter 13 expect to rise across the book-to-market columns.

4.4 Fail-closed panel validation

The analytical grain is exactly one *portfolio-month*: one row per portfolio per calendar month, $25 \times 1{,}193 = 29{,}825$ rows in the canonical panel. A loader is *fail-closed* when it raises an error rather than silently proceeding on any violation of that grain, so that a corrupt panel can never reach a regression. Six checks are enforced, each guarding against a specific way inference can be quietly invalidated:

- **Required columns present** — a missing field (say the risk-free rate) would otherwise be filled with defaults or `NaN` downstream.
- **No null required values** — a single `NaN` propagates through a regression and can silently drop observations.
- **No duplicate (portfolio, date) keys** — duplicates double-count months and corrupt standard errors, likelihoods and ranks.
- **Balanced panel** — every month must contain all 25 portfolios, because the joint HAC covariance and the common-date bootstrap require a rectangular time-by-portfolio array.
- **No missing calendar month** — a gap in the monthly sequence would break the lag structure of the HAC estimator (§2.6).
- **Within-month factor agreement** — the factor values are common to all portfolios in a month, so any within-month disagreement signals a bad join.

```
REQUIRED = {"date", "portfolio", "return", "mkt_rf", "smb", "hml", "rf"}

def validate_panel(df, expected_portfolios=25):
    missing = REQUIRED - set(df.columns)
    if missing:
        raise ValueError(f"Missing columns: {sorted(missing)}")
    if df[list(REQUIRED)].isna().any().any():
        raise ValueError("Null required values")
    if df.duplicated(["portfolio", "date"]).any():
        raise ValueError("Duplicate portfolio-month keys")

    counts = df.groupby("date")["portfolio"].nunique()
    if not counts.eq(expected_portfolios).all():
        raise ValueError("Unbalanced panel")

    dates = pd.PeriodIndex(sorted(df["date"].unique()), freq="M")
    expected = pd.period_range(dates.min(), dates.max(), freq="M")
    if not dates.equals(expected):
        raise ValueError("Missing calendar month")

    factor_spread = df.groupby("date")[["mkt_rf","smb","hml","rf"]].nunique()
    if factor_spread.to_numpy().max() != 1:
        raise ValueError("Factor values disagree within month")
```

A many-to-many join is not more data. If the same portfolio-month appears more than once after a merge, standard errors, likelihoods and rankings can all become invalid. Always validate the analytical key immediately after joining.

4.5 Provenance and the replication/reconciliation distinction

Reproducibility requires knowing exactly which bytes produced a result. For each input the pipeline records the source URL, the acquisition date if known, the sample end date, the file size, a SHA-256 content hash, the weighting block used, the units, and the parser version. A SHA-256 digest is a fixed-length fingerprint of the file's bytes: identical files share a digest, and any change almost certainly changes it, so a recorded hash lets a later reader confirm they hold the same file. Note that a sample *ending* in November 2025 does not prove the file was *downloaded* in November 2025 — the coverage date and the acquisition vintage are different facts.

```
import hashlib

def sha256(path):
    h = hashlib.sha256()
    with open(path, "rb") as f:
        for chunk in iter(lambda: f.read(1 << 20), b""):
            h.update(chunk)
    return h.hexdigest()
```

Replication is not reconciliation. Two different guarantees are often confused. *Numerical replication* asks: given *these* bytes, does an independent implementation reproduce the reported coefficients? Here it does, to a tolerance of 10^{-12} against `statsmodels` (Chapter 15). *Source reconciliation* asks: are *these* bytes the same as the official public source? That is a separate question, and it does not pass.

Paper-specific limitation. The fixed local inputs and the current (May 2026) official snapshot from the Kenneth R. French Data Library differ over their overlapping period — by up to 1.4167 percentage points for portfolio returns and 0.34 percentage points for the FF3/RF fields. Official historical revisions are a plausible but unproven cause. Because the local acquisition vintage is unverified, exact source equality cannot be asserted and is classified NOT_COMPARABLE rather than PASS — a visible, non-blocking finding, not a silent one — even though every within-pipeline integrity check above passes. The local inputs remain immutable and are identified by hash, so the analysis is fully reproducible even while its reconciliation to the live source remains open (Chapter 15).

Chapter 5

Factor regression and the meaning of alpha

This chapter defines the central estimand of the study — the factor alpha — and derives, applies, and interprets the estimator that produces it. The general least-squares and influence-weight results were established in Chapter 2 (§2.4); here they are specialised to the three-factor model, extended to the sampling distribution of alpha, and given an economic reading.

For portfolio i in month t, the Fama–French three-factor (FF3) regression is

$$y_{it} = \alpha_i + \beta_{i,M}\,(\mathrm{Mkt}-\mathrm{RF})_t + \beta_{i,S}\,\mathrm{SMB}_t + \beta_{i,H}\,\mathrm{HML}_t + \varepsilon_{it}, \tag{5.1}$$

where $y_{it} = r_{it} - r_{ft}$ is the portfolio's excess return and the three regressors are the traded Fama–French factors introduced in §2.14.1 and constructed in Chapter 4:

- $(\mathrm{Mkt}-\mathrm{RF})_t$ is the aggregate market return in excess of the risk-free rate;
- SMB_t ("small minus big") is the size factor — the return of small-capitalisation minus large-capitalisation portfolios;
- HML_t ("high minus low") is the value factor — the return of high book-to-market (value) minus low book-to-market (growth) portfolios.

The coefficients $\beta_{i,M}, \beta_{i,S}, \beta_{i,H}$ are the portfolio's sensitivities (loadings) to these factors, α_i is the intercept, and ε_{it} is the month-t residual. Collecting a constant and the $K = 3$ factor series into the design matrix X and writing $\theta_i = (\alpha_i, \beta_i^\top)^\top$, ordinary least squares chooses the intercept and loadings that minimise the residual sum of squares, giving

$$\hat{\theta}_i = (X^\top X)^{-1} X^\top y_i, \qquad \hat{\theta}_i = (\hat{\alpha}_i, \hat{\beta}_i^\top)^\top. \tag{5.2}$$

5.1 Deriving ordinary least squares

This is the specialisation of the general derivation in §2.4; it is repeated here in the factor-model notation because every later step builds on it. For one portfolio, write $y = X\theta + \varepsilon$, where the first column of X is a column of ones and the remaining columns hold the factor returns. The least-squares objective is

$$S(\theta) = (y - X\theta)^\top (y - X\theta) = y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X\theta. \tag{5.3}$$

Differentiating and setting the gradient to zero gives the normal equations,

$$\frac{\partial S}{\partial \theta} = -2X^\top y + 2X^\top X\theta = 0 \implies X^\top X\hat{\theta} = X^\top y \implies \hat{\theta} = (X^\top X)^{-1} X^\top y, \tag{5.4}$$

and the Hessian $2X^\top X \succeq 0$ confirms a global minimum. The residual vector is $e = y - X\hat{\theta}$, and the normal equations imply $X^\top e = 0$: the residuals are orthogonal to the intercept and to every included factor *in the estimation sample*. Orthogonality in-sample is a mechanical property of OLS, not evidence that the model is correct.

5.2 Deriving the intercept influence weights

Substituting $y = X\theta + \varepsilon$ into the estimator gives the estimation error

$$\hat{\theta} - \theta = (X^\top X)^{-1} X^\top \varepsilon. \tag{5.5}$$

Let $q_0 = (1, 0, \ldots, 0)^\top$ select the intercept. Then

$$\hat{\alpha} - \alpha = q_0^\top (X^\top X)^{-1} X^\top \varepsilon = a^\top \varepsilon = \sum_{t=1}^{T} a_t\, \varepsilon_t, \qquad a^\top = q_0^\top (X^\top X)^{-1} X^\top, \tag{5.6}$$

so a is a fixed $T \times 1$ vector of *influence weights* that depends only on the design X, not on the returns. Two facts follow that matter for everything downstream. First, the intercept error is a weighted sum of the residual shocks, which is the object whose variance must be estimated (next section). Second, because all 25 portfolios share the same factors, they share the *same* weights a; applying them to the same-month residual vector turns 25 separate errors into one jointly distributed vector — the bridge from OLS to the joint HAC covariance of Chapter 6 (§2.6).

5.3 The sampling distribution and reported statistics of alpha

A point estimate is meaningless without its uncertainty. From $\hat{\alpha} - \alpha = a^\top \varepsilon$ and the covariance rule (2.6), the variance of the estimated alpha is

$$\mathrm{Var}(\hat{\alpha}) = a^\top \Omega\, a, \qquad \Omega = \mathrm{Cov}(\varepsilon \mid X). \tag{5.7}$$

Three cases make the content of (5.7) concrete.

Classical case. If the residuals were homoskedastic and serially independent, $\Omega = \sigma^2 I$, then $\mathrm{Var}(\hat{\alpha}) = \sigma^2\, a^\top a = \sigma^2\, [(X^\top X)^{-1}]_{00}$. In the centred single-factor toy below, $a_t = 1/T$, so this is simply σ^2/T — the familiar standard error of a sample mean.

Realistic case (HAC). Monthly return residuals are heteroskedastic and autocorrelated (Chapter 13 rejects the classical assumptions), so the one-period formula understates uncertainty. Treating the scalar score $s_t = a_t \varepsilon_t$ as a serially dependent series, the heteroskedasticity- and autocorrelation-consistent variance is the long-run variance of §2.6 with Bartlett weights,

$$\widehat{\mathrm{Var}}_{\mathrm{HAC}}(\hat{\alpha}) = \frac{T}{T-P} \Big\{ \hat{\gamma}_0 + 2 \sum_{l=1}^{L} \Big(1 - \tfrac{l}{L+1}\Big) \hat{\gamma}_l \Big\}, \qquad \hat{\gamma}_l = \sum_{t=l+1}^{T} s_t s_{t-l}, \tag{5.8}$$

with $P = K + 1$ and $L = 12$. This is exactly the diagonal element of the joint covariance of Chapter 6, and reproducing it against `statsmodels` is the single-portfolio audit performed there.

Test statistics. The HAC standard error is $\mathrm{SE}_{\mathrm{HAC}}(\hat{\alpha}) = \sqrt{\widehat{\mathrm{Var}}_{\mathrm{HAC}}(\hat{\alpha})}$, and the individual test of $H_0 : \alpha_i = 0$ uses

$$t_i = \frac{\hat{\alpha}_i}{\mathrm{SE}_{\mathrm{HAC}}(\hat{\alpha}_i)}, \qquad p_i = 2\,\Phi(-|t_i|), \tag{5.9}$$

a large-sample normal reference; the 25 resulting p-values are then false-discovery-controlled (§2.8). Two goodness-of-fit summaries accompany each fit: the coefficient of determination

$R^2 = 1 - \text{SSE}/\text{SST}$ (the share of the portfolio's return variance explained by the factors) and the residual volatility $\hat{\sigma}_\varepsilon = \text{sd}(e)$.

A standard error is not a return volatility. The alpha standard error measures how much $\hat{\alpha}$ would move across resamples of the history; the residual volatility $\hat{\sigma}_\varepsilon$ measures how much the portfolio's monthly abnormal return scatters around the fit. They are different quantities with different units of interpretation, and conflating them is a frequent error: a portfolio can have a precisely estimated alpha (small SE) yet volatile residuals (large $\hat{\sigma}_\varepsilon$), or vice versa.

5.4 Interpretation

- β**:** sensitivity of the portfolio to a factor, conditional on the other included factors.
- α**:** average residual return after the included linear factor exposures are removed.
- ε**:** month-specific residual, containing noise, omitted factors, nonlinearities, changing exposures and measurement error.

Alpha is not automatically skill. For passive characteristic portfolios, a non-zero alpha is better described as a pricing error or benchmark-relative residual mean. The benchmark may be incomplete, so the same interpretive caution as in §3.2 applies: $\hat{\alpha}$ measures return unexplained by *these* factors, not by every source of risk.

What "conditional on the other factors" means. The loading $\beta_{i,H}$ on HML is not the simple regression slope of y_i on HML; it is a *partial* slope. By the Frisch–Waugh–Lovell theorem, $\hat{\beta}_{i,H}$ equals the slope from regressing y_i on the residual of HML after HML has itself been projected on the constant and the other factors $(\text{Mkt}-\text{RF}, \text{SMB})$. Equivalently, partialling the other regressors out of both sides isolates the sensitivity to the part of HML that is orthogonal to the remaining factors. This is why a loading can change — even flip sign — when a factor is added or removed, and it is the mechanism behind the benchmark dependence of alpha discussed in §3.3: moving from FF3 to FF5 re-partials every loading and therefore re-defines the intercept.

The constant-beta assumption. Within one estimation window the loadings β_i are treated as constant. If the true exposures drift over the window, the drift is absorbed into the residual ε_{it}, where it can inflate residual dependence or masquerade as alpha. This is one reason the study estimates on rolling windows (Chapter 10) and relies on HAC inference rather than the classical variance: the linear, constant-β model is a local approximation, not a literal truth.

5.5 Complete numerical toy regression

Use five months, one centred factor f=(-2,-1,0,1,2)$'$, and three portfolios. All numbers below are monthly percentages solely to make the arithmetic readable.

Month	f	Portfolio A y	Portfolio B y	Portfolio C y
1	-2	-1.70	-0.83	-2.55
2	-1	-1.00	-0.58	-1.26
3	0	0.40	0.32	-0.08
4	1	1.00	0.42	1.14
5	2	2.30	1.17	2.25

Here $X = [\mathbf{1},\ f\,]$. Because $\sum_t f_t = 0$ and $\sum_t f_t^2 = 10$,

$$X^\top X = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}, \qquad (X^\top X)^{-1} = \begin{bmatrix} 1/5 & 0 \\ 0 & 1/10 \end{bmatrix}. \tag{5.10}$$

The orthogonality of the constant and the centred factor diagonalises $X^\top X$, so the intercept and slope decouple: $\hat{\alpha}$ is simply the sample mean $\frac{1}{5}\sum_t y_t$, and $\hat{\beta} = (\sum_t f_t y_t)/(\sum_t f_t^2)$. The fitted results are

Portfolio	$\hat{\alpha}$ monthly	$\hat{\beta}$	Linear annualised α	Residual vector
A	0.20%	1.00	240 bps	(0.10,-0.20,0.20,-0.20,0.10)
B	0.10%	0.50	120 bps	(0.07,-0.18,0.22,-0.18,0.07)
C	-0.10%	1.20	-120 bps	(-0.05,0.04,0.02,0.04,-0.05)

Since the factor is centred, a_t=1/5 for every t. The residual vectors are individually orthogonal to 1 and f, but they co-move across portfolios. That same-date co-movement is exactly what the full joint covariance retains.

5.6 Reference implementation

```
import numpy as np
import statsmodels.api as sm

def fit_factor_model(y, factors, hac_lags=12):
    X = sm.add_constant(np.asarray(factors, float))
    model = sm.OLS(np.asarray(y, float), X).fit()
    robust = model.get_robustcov_results(
        cov_type="HAC", maxlags=hac_lags, use_correction=True
    )
    return {
        "alpha_monthly": model.params[0],
        "alpha_bps_year": model.params[0] * 12 * 10_000,
        "betas": model.params[1:],
        "residuals": model.resid,
        "hac_se_alpha": robust.bse[0],
        "hac_p_alpha": robust.pvalues[0],
    }
```

Independent replication means comparing the canonical estimator with a separately implemented calculation such as statsmodels. Two functions calling the same internal routine are not independent replication.

Chapter 6

Full cross-portfolio Newey-West covariance

This chapter builds the object on which all joint inference rests: the full 25×25 covariance of the estimated alpha vector. The general long-run-variance and Newey–West theory was derived in §2.6; here we motivate why an ordinary — or even a portfolio-by-portfolio — covariance is inadequate for this problem, assemble the joint matrix from the influence scores of Chapter 5, and audit it numerically.

6.1 Why ordinary OLS covariance is inadequate

The classical OLS variance $\sigma^2(X^\top X)^{-1}$ rests on residuals that are homoskedastic, serially independent, and (across portfolios) mutually uncorrelated. Monthly factor-model residuals violate all three, so the classical variance is the wrong uncertainty object. The general form is the sandwich of §2.4, $\mathrm{Var}(\hat{\theta}) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}$, in which the "bread" $(X^\top X)^{-1}$ is fixed but the "meat" $X^\top \Omega X$ must be estimated without assuming $\Omega = \sigma^2 I$.

6.1.1 Heteroskedasticity

The conditional residual variance $\mathrm{Var}(\varepsilon_{it} \mid X)$ changes over time — volatility clusters, with calm and turbulent regimes. If the true $\mathrm{Var}(\varepsilon_{it}) = \sigma_t^2$ varies but the classical formula plugs in a single $\hat{\sigma}^2$, the reported standard error is biased, and the direction of the bias depends on whether large residuals coincide with high-leverage months. The ARCH-LM test rejects constant conditional variance for all 25 portfolios (Chapter 13).

6.1.2 Autocorrelation

Residual shocks remain correlated across nearby months, so $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t-\ell}) \neq 0$ for small ℓ. The classical variance uses only the contemporaneous term and ignores these lag covariances; the correct object is the *long-run* variance (2.19), which sums the autocovariances across lags. Ignoring positive serial dependence understates the true sampling variability. The Ljung–Box test rejects the no-autocorrelation null for many portfolios (Chapter 13), and the Newey–West estimator includes lags to $L = 12$.

6.1.3 Cross-sectional dependence

Different portfolios are hit by the same monthly shocks — a common market move affects all 25 at once — so $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{jt}) \neq 0$ across $i \neq j$. This is dependence *across the cross-section within a* *date*, invisible to any single-portfolio calculation. It is precisely the off-diagonal information that the joint test of Chapter 7 and the multivariate shrinkage of Chapter 8 consume: the variance of a difference $\hat{\alpha}_i - \hat{\alpha}_j$ is $V_{ii} + V_{jj} - 2V_{ij}$, and dropping V_{ij} misstates every pairwise comparison.

Twenty-five separate HAC standard errors preserve heteroskedasticity and serial dependence *within* each portfolio — they give the 25 diagonal entries of the covariance — but discard the cross-portfolio covariances. The construction below recovers the full matrix.

6.2 Influence-score construction

Because every portfolio uses the same factor matrix X, the OLS intercept estimation error is a weighted sum of residuals with weights common to all portfolios (§5.2). Let a_t be the intercept weight for month t, e_t the 25-vector of same-month residuals, and $g_t = a_t e_t$ the 25-vector alpha score. The lag-ℓ score cross-product and the Bartlett weights are

$$\Gamma_\ell = \sum_{t=\ell+1}^{T} g_t\, g_{t-\ell}^\top, \qquad w_\ell = 1 - \frac{\ell}{L+1}, \tag{6.1}$$

and the joint HAC covariance is

$$\hat{V}_{\text{HAC}} = c\Big\{\Gamma_0 + \sum_{\ell=1}^{L} w_\ell\,(\Gamma_\ell + \Gamma_\ell^\top)\Big\}, \tag{6.2}$$

with $L = 12$ lags and the finite-sample correction $c = T/(T-P)$, $P = K+1$. This is the applied form of the general estimator (2.20) and its joint specialisation (2.22) derived in §2.6; the Bartlett weights decline linearly with lag and guarantee that $\hat{V}_{\text{HAC}}$ is positive semi-definite (the Fejér-kernel argument there).

6.3 Deriving the joint covariance from first principles

The intercept influence weight is itself $a_t = q_0^\top (X^\top X)^{-1} x_t$, where $q_0 = (1, 0, \ldots, 0)^\top$ selects the intercept and $x_t^\top$ is row t of X (§5.2); it is a fixed scalar depending only on the common design, not on any portfolio's returns. Stack the N portfolio residuals at month t into $e_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})^\top$. Because the same a_t multiplies every portfolio, the N-vector of alpha-estimation errors and its per-month score are

$$\hat{\alpha} - \alpha = \sum_{t=1}^{T} a_t e_t = \sum_{t=1}^{T} g_t, \qquad g_t = a_t e_t. \tag{6.3}$$

Without assuming independence through time, the exact covariance is the full double sum

$$\operatorname{Var}(\hat{\alpha} - \alpha) = \sum_{t=1}^{T}\sum_{s=1}^{T} \operatorname{Cov}(g_t, g_s). \tag{6.4}$$

Grouping by lag. The double sum has T^2 terms indexed by (t, s); re-index them by the lag $\ell = t - s$. Under weak stationarity the covariance depends on ℓ alone, so define $\Gamma_\ell = \operatorname{Cov}(g_t, g_{t-\ell}) = \mathbb{E}[g_t g_{t-\ell}^\top]$. The T diagonal terms ($\ell = 0$) contribute Γ_0. For each $\ell \geq 1$, the pairs with $t - s = \ell$ contribute Γ_ℓ, while the pairs with $s - t = \ell$ contribute $\operatorname{Cov}(g_t, g_{t+\ell}) = \Gamma_{-\ell} = \Gamma_\ell^\top$ (covariance of a stationary series satisfies $\Gamma_{-\ell} = \Gamma_\ell^\top$). Collecting the two directions, the double sum collapses to the one-sided lag representation

$$\operatorname{Var}(\hat{\alpha} - \alpha) = \Gamma_0 + \sum_{\ell=1}^{T-1} (\Gamma_\ell + \Gamma_\ell^\top), \tag{6.5}$$

which is exactly the long-run covariance (2.19). The transpose term is not decoration: it supplies the negative-lag contributions and makes the matrix symmetric.

From population to estimator. Three finite-sample steps turn the population identity (6.5) into the usable estimator (6.2). First, each population Γ_ℓ is replaced by its sample cross-product $\hat{\Gamma}_\ell = \sum_{t=\ell+1}^{T} g_t g_{t-\ell}^\top$ from (6.1). Second, the sum is *truncated* at a bandwidth $L \ll T$, because $\hat{\Gamma}_\ell$ at large ℓ is built from only $T - \ell$ products and is too noisy to include. Third, each retained lag is *tapered* by the Bartlett weight $w_\ell = 1 - \ell/(L+1)$. Element (i, j) of the result estimates $\mathrm{Cov}(\hat{\alpha}_i, \hat{\alpha}_j)$, and setting $i = j$ recovers the single-portfolio HAC variance (5.8) — the identity the numerical audit below exploits.

Object	Dimension	Role
X	T × (K+1)	Common regression design
a	T × 1	Intercept influence weights
$\mathrm{e_t}$	N × 1	Same-month residual vector
$\mathrm{g_t{=}a_te_t}$	N × 1	Same-month alpha score
Γ_ℓ	N × N	Lag-ℓ score cross-product
$\hat{V}_{\mathrm{HAC}}$	N × N	Joint alpha-estimation covariance

6.4 Tapering, lag length, and the finite-sample correction

Three choices in (6.2) deserve explanation, because they control whether the estimate is a valid covariance and how much it can be trusted.

Why taper (positive semi-definiteness). A truncated *un*weighted sum $\Gamma_0 + \sum_{\ell=1}^{L}(\Gamma_\ell + \Gamma_\ell^\top)$ — a rectangular window — need not be positive semi-definite, and a "covariance" with a negative eigenvalue is meaningless (it would imply a portfolio combination of alphas with negative variance). The Bartlett taper fixes this. The weighted sum can be written as a $T^{-1} \sum_{|\ell| \le L}(1 - |\ell|/(L + 1)) \sum_t g_t g_{t-\ell}^\top$, which is a *Fejér-kernel* (triangular) smoothing of the periodogram; the Fejér kernel is non-negative, so the resulting matrix is guaranteed positive semi-definite for any data (§2.6). This is why the linearly declining weights $w_\ell = 1 - \ell/(L + 1)$ are used rather than a sharp cutoff.

Choosing the bandwidth L. The lag length trades bias against variance. Too small an L omits genuine dependence beyond lag L and biases the variance downward when autocorrelation is positive and persistent; too large an L admits many noisy $\hat{\Gamma}_\ell$ and inflates the estimator's own sampling variability. Consistency of the HAC estimator requires $L \to \infty$ but $L/T \to 0$ as $T \to \infty$, so the bandwidth must grow with the sample yet slowly. The canonical monthly regressions fix $L = 12$: one year of lags, a standard convention that lets residual dependence persist up to twelve months while keeping each $\hat{\Gamma}_\ell$ well estimated on $T = 1{,}193$ months. Where the series is short — the annual forward-validation statistics of Chapter 11 — a fixed $L = 12$ would be too large, so the automatic Newey–West (1994) rule $L = \lfloor 4(T/100)^{2/9} \rfloor$ is used instead (§2.6).

The finite-sample correction $c = T/(T - P)$. Residuals are computed from an estimated $\hat{\theta}$, which consumes $P = K + 1$ degrees of freedom and makes the raw residual cross-products slightly too small on average, exactly as dividing by T rather than $T - P$ understates the OLS residual variance. Multiplying by $c = T/(T - P)$ is the multivariate analogue of that degrees-of-freedom correction and removes the leading small-sample downward bias; with $T = 1{,}193$ and $P = 4$ it is $c \approx 1.003$, negligible here but retained for exactness and reported in the manifest.

6.5 Toy covariance calculation

Reusing the five-month example of Chapter 5, the centred factor gives $a_t = 0.2$ for every t. Ignoring lag terms only for illustration, the residual cross-product matrix across the three portfolios A, B, C is

$$E^\top E = \begin{bmatrix} 0.140 & 0.130 & -0.022 \\ 0.130 & 0.123 & -0.017 \\ -0.022 & -0.017 & 0.0086 \end{bmatrix}. \tag{6.6}$$

Thus the contemporaneous score cross-product is $\Gamma_0 = 0.2^2\, E^\top E$, and with $P = 2$ regression parameters the finite-sample factor is $c = T/(T-P) = 5/3$, so the illustrative covariance is $\frac{5}{3} \times 0.04 \times E^\top E$. The positive A–B entry means the two portfolios' alpha-estimation errors tend to move together; treating them as independent would discard exactly this cross-portfolio information.

6.6 Numerical audit

A HAC covariance is only useful if it is genuinely a covariance. Bartlett weights guarantee positive semi-definiteness in theory (§2.6), but finite-precision arithmetic can introduce tiny asymmetries and negative eigenvalues, so the estimate is checked before use: it is symmetrised, its eigenvalues are inspected, and only negligible, scale-relative negatives are floored. A materially indefinite result *raises* rather than being silently regularised, so numerical repair is always visible in the run manifest.

```
V = 0.5 * (V + V.T)                    # remove floating asymmetry
eig = np.linalg.eigvalsh(V)
scale = max(np.linalg.norm(V, 2), 1.0)
tol = 1e-12 * scale
if eig.min() < -tol:
    raise ArithmeticError("Materially indefinite HAC covariance")
condition_number = np.linalg.cond(V)
# Record any tiny numerical flooring; never hide large regularisation.
```

Paper result. The canonical 25×25 matrix is positive definite without any flooring and has condition number 139.4; the ratio of largest to smallest eigenvalue is finite and manageable, so the linear solves behind the Wald statistic (Chapter 7) and the empirical-Bayes posterior (Chapter 8) are numerically stable. A large condition number would signal near-collinearity of the alpha errors and unstable inversion; 139.4 is comfortably below that regime.

Independent replication. The diagonal of $\hat{V}_{\mathrm{HAC}}$ must equal the 25 single-portfolio HAC variances (5.8). Recomputing each portfolio's alpha and HAC standard error with an independent implementation (`statsmodels`) reproduces the diagonal to a tolerance of 10^{-12} for all 25 portfolios. This checks the joint construction against a separate codebase; two functions calling the same internal routine would not constitute independent replication. Off-diagonal entries have no single-portfolio analogue and are validated instead through the positive-definiteness and symmetry checks above.

6.7 What HAC does and does not do

It is worth stating the estimator's remit precisely, because HAC is easy to over-trust. It makes the *covariance estimate* robust to the forms of dependence it models — heteroskedasticity, serial correlation to lag L, and contemporaneous cross-portfolio correlation — and thereby makes the joint Wald test of Chapter 7 asymptotically valid without an IID or Gaussian assumption. It does none of the following: it does not make the factor model correct, does not recover omitted priced factors or repair time-varying betas, does not capture serial dependence beyond lag L, and does not fix data errors such as a bad join or the wrong units. HAC is a statement about the *sampling distribution of the estimator*, not about the *truth of the model*; a portfolio can have a perfectly estimated HAC standard error around an alpha that is entirely an artefact of an incomplete benchmark.

HAC helps with	HAC does not repair
Heteroskedasticity and finite-lag serial dependence in covariance estimation	Omitted factors, nonlinear payoffs or time-varying betas
Cross-portfolio covariance when scores are stacked by common date	Bad joins, wrong units or an incorrect data vintage
Asymptotically robust Wald inference	Gaussian likelihood assumptions used elsewhere

Chapter 7

Joint testing and false-discovery control

7.1 The joint HAC/Wald test

The null hypothesis is H_0: $\alpha = 0$ for all 25 portfolios. The dependence-robust Wald statistic is:

$$W = \hat{\alpha}'\hat{V}_{\mathrm{HAC}}^{-1}\hat{\alpha} \approx \chi^2_{25} \quad \text{under } H_0 \tag{7.1}$$

Use a linear solve rather than forming an explicit inverse:

```
W = float(alpha @ np.linalg.solve(V_hac, alpha))
p = scipy.stats.chi2.sf(W, df=len(alpha))
```

Correct conclusion. Rejecting joint zero alpha means at least some elements of the alpha vector are inconsistent with zero under the model. It does not say the alphas are positive, the ranking is stable, or a portfolio displays managerial skill.

7.2 The conventional GRS test

$$GRS = [(T - N - K)/N] \cdot [1 + \bar{f}'\hat{\Omega}_f^{-1}\bar{f}]^{-1} \cdot \hat{\alpha}'\hat{\Sigma}_\varepsilon^{-1}\hat{\alpha} \sim F_{N,T-N-K} \tag{7.2}$$

Here $\hat{\Sigma}_\varepsilon$ is the N × N contemporaneous residual covariance, $\hat{\Omega}_{\mathrm{f}}$ is the K × K factor covariance, and $\bar{f}$ is the sample mean factor vector. The quadratic form $\hat{\alpha}'\hat{\Sigma}_\varepsilon{}^{-1}\hat{\alpha}$ asks whether alpha is large relative to residual risk. The factor term discounts this evidence when the factors themselves have a large sample mean relative to their covariance.

Why the F distribution? Under a correctly specified linear factor model with residual vectors that are independent through time, jointly Gaussian, homoskedastic and independent of factor realisations, the appropriately scaled alpha quadratic form can be written as a ratio of independent quadratic forms. That ratio has an exact $\mathrm{F}_{\mathrm{N,T-N-K}}$ distribution. Serial dependence, conditional heteroskedasticity or non-normality breaks the exact finite-sample argument.

Assumption ledger. GRS is not "wrong"; it answers a sharper question under stronger assumptions. In this study, residual diagnostics reject several of those assumptions, so GRS is a conventional sensitivity check and the HAC/Wald statistic is the primary joint inference.

7.3 Why 25 individual p-values need correction

If 25 true null hypotheses are each tested at 5%, false positives become likely. The Benjamini-Hochberg (BH) procedure controls the expected proportion of false discoveries among the rejected hypotheses under its stated dependence conditions.

1. Sort p-values: $p_{(1)} \leq \ldots \leq p_{(m)}$.
2. Find the largest k satisfying $p_{(k)} \leq kq/m$.
3. Reject hypotheses 1 through k.

7.4 A complete BH example

Suppose m = 5 tests produce p-values (0.003, 0.011, 0.018, 0.040, 0.230), already sorted, and the target FDR is q = 0.05. The step-up thresholds are:

Order i	$p_{(i)}$	i q/m	Passes?
1	0.003	0.010	Yes
2	0.011	0.020	Yes
3	0.018	0.030	Yes
4	0.040	0.040	Yes
5	0.230	0.050	No

The largest passing order is k = 4, so the first four nulls are rejected. Notice the step-up logic: one does not stop at the first local failure. The adjusted q-values follow from

$$q_{(i)} = \min_{j \geq i} mp_{(j)}/j, \quad \text{truncated at 1 and mapped back to the original order.} \tag{7.3}$$

BH controls FDR under independence and under broad positive-dependence conditions. Portfolio tests are correlated, so the full dependence structure should still be reported and the discovery family must be stated explicitly. A more conservative arbitrary-dependence procedure exists, but needlessly replacing BH without a design argument can destroy power.

```
def bh_qvalues(p):
    p = np.asarray(p, float)
    m = p.size
    order = np.argsort(p)
    ranked = p[order]
    adjusted = ranked * m / np.arange(1, m + 1)
    adjusted = np.minimum.accumulate(adjusted[::-1])[::-1]
    q = np.empty(m)
    q[order] = np.clip(adjusted, 0, 1)
    return q
```

Keep separate testing families separate. The 25 raw-alpha tests and 25 SFA boundary tests answer different questions. Correct each family separately and report the family definition.

In the study, five raw alphas survive 5% BH control, but four are significantly negative. "Significant" is therefore not synonymous with "good."

Chapter 8

Multivariate empirical Bayes and partial pooling

8.1 The winner's curse

When the largest estimate is selected from 25 noisy estimates, it is likely to be unusually high partly because of noise. Partial pooling shrinks estimates toward a cross-sectional centre, with the amount of shrinkage determined by sampling uncertainty and the estimated dispersion of true alphas.

$$\hat{\alpha}|\alpha \sim N(\alpha, \hat{V}_{\text{HAC}}), \alpha \sim N(\mu 1, \tau^2 I) \tag{8.1}$$

$$E(\alpha|data) = \mu 1 + \tau^2(\hat{V}_{\text{HAC}} + \tau^2 I)^{-1}(\hat{\alpha} - \mu 1) \tag{8.2}$$

$$Var(\alpha|data) = \tau^2 I - \tau^4(\hat{V}_{\text{HAC}} + \tau^2 I)^{-1} \tag{8.3}$$

With a diagonal V, each portfolio shrinks independently. With the full V, a portfolio's posterior update also reflects how its alpha-estimation error co-moves with the others.

8.2 Deriving the posterior by completing the square

Write a = $\hat{\alpha}$ and let 1 denote the N-vector of ones. Ignoring constants that do not depend on α, Bayes' rule gives:

$$logp(\alpha|a) = -\frac{1}{2}(a-\alpha)'V^{-1}(a-\alpha) - \frac{1}{2}(\alpha - \mu 1)'(\tau^{-2}I)(\alpha - \mu 1) + C. \tag{8.4}$$

Collect the quadratic and linear terms in α:

$$logp(\alpha|a) = -\frac{1}{2}\alpha' P \alpha + \alpha' b + C', P = V^{-1} + \tau^{-2}I, b = V^{-1}a + \tau^{-2}\mu 1. \tag{8.5}$$

Complete the square with $\text{C}_{\text{post}}\text{=P}^{-1}$ and $\text{m}_{\text{post}}\text{=C}_{\text{post}}\text{b}$:

$$-\frac{1}{2}\alpha' P \alpha + \alpha' b = -\frac{1}{2}(\alpha - m_{\text{post}})' P (\alpha - m_{\text{post}}) + \frac{1}{2} m_{\text{post}}{}' P m_{\text{post}}. \tag{8.6}$$

Therefore $\alpha|$a is multivariate normal. Matrix identities turn the precision-form solution into the update form shown above:

$$m_{\text{post}} = \mu 1 + \tau^2 (V + \tau^2 I)^{-1}(a - \mu 1), \tag{8.7}$$

$$C_{\text{post}} = (V^{-1} + \tau^{-2} I)^{-1} = \tau^2 I - \tau^4 (V + \tau^2 I)^{-1}. \tag{8.8}$$

If V is diagonal with entry s_i^2, the formula reduces to the familiar scalar weighted average:

$$E(\alpha_i|a_i) = w_i a_i + (1 - w_i)\mu, w_i = \tau^2/(\tau^2 + s_i^2). \tag{8.9}$$

For example, with $\mu = -26$ bps/year, $\tau = 94$ bps/year, $a_i = 250$ bps/year and $s_i = 150$ bps/year,

$$w_i = 94^2/(94^2 + 150^2) = 0.282, \quad \text{posterior mean} = 0.282(250) + 0.718(-26) = 51.8 \text{ bps/year.} \tag{8.10}$$

The estimate shrinks sharply because its sampling uncertainty exceeds the cross-sectional prior dispersion. In the actual multivariate calculation there is no independent scalar weight for each portfolio: off-diagonal elements of V couple all updates.

8.3 Estimating μ and τ by marginal likelihood

The marginal model is $\hat{\alpha}$ ~ N(μ1, V + τ^2I). Profile marginal maximum likelihood searches over $\tau \geq 0$, solves for μ conditional on τ, and selects the pair that maximises the marginal likelihood.

Let S(τ)=V+τ^2I. The marginal log-likelihood is

$$\ell(\mu, \tau) = -\frac{1}{2}[N log(2\pi) + log|S(\tau)| + (a - \mu 1)' S(\tau)^{-1}(a - \mu 1)]. \tag{8.11}$$

Differentiating with respect to μ gives

$$\partial \ell / \partial \mu = 1' S^{-1}(a - \mu 1). \tag{8.12}$$

Setting this to zero yields the generalised least-squares profile estimate:

$$\hat{\mu}(\tau) = [1' S(\tau)^{-1} a]/[1' S(\tau)^{-1} 1]. \tag{8.13}$$

Substitute $\hat{\mu}(\tau)$ back into ℓ and optimise the resulting one-dimensional function over $\tau \geq 0$. Optimising log τ enforces positivity, but the $\tau = 0$ boundary must be evaluated explicitly rather than excluded by the transformation.

```
def profiled_mu(alpha_hat, V, tau2):
    S = V + tau2 * np.eye(len(alpha_hat))
    one = np.ones(len(alpha_hat))
    Sinv_one = np.linalg.solve(S, one)
    return (Sinv_one @ alpha_hat) / (Sinv_one @ one)

def neg_profile_loglik(log_tau, alpha_hat, V):
    tau2 = np.exp(2 * log_tau)
```

```
S = V + tau2 * np.eye(len(alpha_hat))
mu = profiled_mu(alpha_hat, V, tau2)
d = alpha_hat - mu
sign, logdet = np.linalg.slogdet(S)
if sign <= 0:
    return np.inf
return 0.5 * (logdet + d @ np.linalg.solve(S, d))
```

8.4 Raw tests and posterior intervals answer different questions

Object	Question	Uses
Raw alpha p/q	Is the unpooled regression intercept distinguishable from zero?	Testing and discovery control
Posterior mean	What is the partially pooled estimate under the hierarchical model?	Uncertainty-aware scoring
Posterior interval	What range is plausible under the fitted hierarchical model?	Magnitude uncertainty
Bootstrap rank interval	Where might the portfolio rank under historical resampling?	Ranking uncertainty

Paper result. The fitted prior centre is -26 annualised bps and prior dispersion is 94 bps. Partial pooling changes at least one portfolio by ten rank positions. Posterior intervals remain approximate because τ uncertainty is not fully integrated and V is treated as known during the conditional posterior calculation.

Chapter 9

Common-date circular block bootstrap for rank uncertainty

9.1 Why a rank needs its own uncertainty analysis

A rank is a discontinuous function of all 25 scores. A standard error for portfolio i alone cannot tell us whether it remains above portfolio j when shared historical shocks are resampled.

9.2 Why blocks and why common dates?

- Monthly observations are dependent; resampling individual months destroys serial structure.
- A 12-month block preserves local dependence across a year.
- Circular wrapping gives every month equal opportunity to begin a complete block.
- The same resampled date indices must be used for every portfolio to preserve cross-sectional shocks.

9.3 Full-refit algorithm

```
for b in range(B):
    idx = circular_block_indices(T, block_length=12, rng=rng)
    y_b = y[idx, :]              # same dates for all portfolios
    f_b = factors[idx, :]

    alpha_b, beta_b, resid_b = fit_all_ols(y_b, f_b)
    V_b = joint_newey_west_alpha_covariance(resid_b, f_b, lags=12)
    mu_b, tau_b = fit_empirical_bayes(alpha_b, V_b)
    post_b = posterior_mean(alpha_b, V_b, mu_b, tau_b)
    rank_b = descending_rank(post_b)
    store(post_b, rank_b)
```

Do not hold the original betas, covariance or hyperparameters fixed inside the bootstrap. That would understate estimator uncertainty.

9.4 A circular-index example

Take T = 10 observations labelled 0,. . . ,9 and block length L = 4. If random starting indices are 8, 3 and 6, the circular blocks are:

$$(8, 9, 0, 1), (3, 4, 5, 6), (6, 7, 8, 9). \tag{9.1}$$

Concatenate and truncate to length T, giving (8,9,0,1,3,4,5,6,6,7). For the portfolio panel, exactly this index vector selects rows of every portfolio and every factor. The duplicated 6 is valid bootstrap sampling; cross-sectional alignment at each selected date is preserved.

Block length trades bias against variance. Very short blocks resemble IID resampling and break dependence; very long blocks leave few effectively distinct resamples. Twelve months is substantively interpretable here, but a defensible workflow also checks nearby lengths. With B draws, the Monte Carlo standard error of an estimated probability $\hat{p}$ is approximately

$$MCSE(\hat{p}) \approx \sqrt{\hat{p}(1-\hat{p})/B}. \tag{9.2}$$

At $\hat{p} = 0.5$ and B = 5,000 this is about 0.0071, or 0.71 percentage points. Quantile endpoints generally need more draws than posterior means, which is why the 1,000-versus-5,000 stability comparison matters.

9.5 Outputs

- Median rank and central 95% rank interval.
- Probability of being in the top or bottom quintile.
- Bootstrap score intervals.
- Stability comparison across 1,000 and 5,000 draws.

Paper result. With 5,000 draws, rank-interval endpoints moved by at most one rank relative to 1,000 draws, but alpha-tail endpoints moved by as much as 27 bps/year. SMALL HiBM is point rank 1 and reaches rank 6 in its central rank interval.

Chapter 10

Rolling persistence, overlap and quintile mobility

10.1 The overlap illusion

Two 120-month windows ending 12 months apart share 108 months, or 90% of their data. High correlation between their rankings is therefore partly mechanical. A 120-month separation produces disjoint decades and is the structurally interpretable comparison.

$$Overlap fraction = max(0, W - h)/W \tag{10.1}$$

For W = 120 and horizon h = 12, the fraction is 0.90; for h = 120, it is zero.

10.2 Persistence metrics

- **Spearman rank correlation:** agreement in order, invariant to monotone score transformations.
- **Pearson score correlation:** linear agreement in estimated posterior scores.
- **Mean absolute rank change:** average number of positions moved.
- **Transition matrix:** probability of moving from one quintile to another.
- **Jaccard overlap:** intersection divided by union for top/bottom membership sets.

$$J(A, B) = |A \cap B|/|A \cup B| \tag{10.2}$$

10.3 Correct interpretation

Horizon	Window overlap	Mean rank correlation	Interpretation
12 months	90%	0.863	Mostly an overlap diagnostic
60 months	50%	0.483	Still contaminated by overlap
120 months	0%	0.156	Weak structural persistence

The same-quintile probability at the disjoint 120-month horizon is about 25%, only modestly above the 20% uniform-mobility benchmark. This warns against treating the point ordering as a stable hierarchy.

Chapter 11

Strictly forward validation

Design an evaluation where every model choice and estimate used for ranking is fixed before the future observations are touched.

11.1 Timeline

120-month training data

Fit betas, V, μ, τ

Freeze scores and ranks

Observe next 12 months

Evaluate future alpha

$$A_i^+ = mean_{\text{future t}}[y_{\text{it}} - \hat{\beta}'_{i,train} f_t] \tag{11.1}$$

The beta is frozen. Re-estimating beta on the evaluation year would allow future data to redefine the target being evaluated.

At cutoff c, let the training set be {c−119,. . . ,c} and the evaluation set be {c+1,. . . ,c+12}. The ranking function R_c must be measurable with respect to information available at c. In probability notation, if $\mathcal{F}_c$ is the information set at the cutoff, then

$$R_c \in \mathcal{F}_c, while A_c^+ \in \mathcal{F}_{c+12} \setminus \mathcal{F}_c. \tag{11.2}$$

This is the formal meaning of "look-ahead free." Every estimated beta, covariance, hyperparameter, scaling rule, tie-breaking rule and portfolio assignment must be determined by $\mathcal{F}_c$.

11.2 Evaluation statistics

1. Spearman correlation between training scores and future alpha across the 25 portfolios.
2. Average future alpha of the training top quintile minus the average future alpha of the training bottom quintile.

For correlations, transform each annual Spearman value using z = atanh(ρ), aggregate z with HAC inference, then transform back with tanh. This prevents averaging correlations on an inappropriate raw scale.

$$z_w = atanh(\rho_w) = \frac{1}{2} log[(1+\rho_w)/(1-\rho_w)], \bar{\rho} = tanh(\bar{z}). \tag{11.3}$$

If z_w is serially dependent across evaluation windows, estimate the long-run variance of its mean with HAC. A 95% interval on the z scale is $\bar{z} \pm 1.96 \cdot \mathrm{SE}_{\mathrm{HAC}}(\bar{z})$, with both endpoints mapped back through tanh.

11.3 Leakage assertions

```
assert train_dates.max() < eval_dates.min()
assert len(set(train_dates) & set(eval_dates)) == 0
assert beta_timestamp < eval_dates.min()
assert rank_timestamp < eval_dates.min()
assert hyperparameter_timestamp < eval_dates.min()
assert preprocessing_fit_timestamp < eval_dates.min()
```

Paper result. Across 89 non-overlapping evaluation years, mean rank correlation is 0.102 with HAC 95% interval [0.055, 0.149]. The top-minus-bottom quintile spread averages 243 bps/year, but its IQR of 498 bps exceeds its mean and negative years remain frequent.

A positive average is not a timing rule. Forward validation shows association using past information. It does not yet prove that annual re-estimation adds value over a simpler stable ranking.

Chapter 12

The incremental test: dynamic forecast or stable ordering?

12.1 The identification problem

Suppose FF3 persistently underprices a fixed set of characteristic cells. Past alpha will then predict future alpha even if annual re-estimation contains no useful time variation. The forward level test cannot distinguish this from dynamic ranking skill.

12.2 Comparator rankings

Comparator	Uses future data?	Purpose
Size-ascending / B/M-descending gradient	No	Naive characteristic ordering
Diagonal value-minus-size gradient	No	Alternative simple characteristic ordering
Expanding-window ranking	No	Fair strong comparator using all data available at each cutoff
Full-sample alpha oracle	**Yes**	Look-ahead upper bound, never a tradable strategy

12.3 Increment definition

$$\Delta_w = performance_w(rollingrank) - performance_w(benchmarkrank) \tag{12.1}$$

Aggregate the per-window increments using a one-sided HAC t-test for $E(\Delta) > 0$ and a block bootstrap. The fair verdict rests on the expanding-window comparator, because both rankings use only information available at the cutoff.

12.4 A numerical example that separates level from increment

Suppose four evaluation years produce rolling-rank correlations (0.12, 0.08, 0.16, 0.04) and expanding-rank correlations (0.15, 0.11, 0.18, 0.07). Both methods predict positively:

$$mean(\rho_{\text{rolling}}) = 0.10, mean(\rho_{\text{expanding}}) = 0.1275. \tag{12.2}$$

But the paired increments are $(-0.03, -0.03, -0.02, -0.03)$, with mean -0.0275. Thus a positive forward level is perfectly compatible with a negative incremental contribution from rolling re-estimation. The shared fixed cross-sectional signal explains the positive level; discarding older observations adds estimation noise.

12.5 Interpretation of the reported evidence

- The rolling rank beats the two naive characteristic gradients.
- It does not beat the expanding-window rank: mean correlation increment -0.042, one-sided $p = 0.97$. For the prespecified alternative $E(\Delta)>0$, such a large p-value means the estimate lies in the wrong direction; it is not evidence that the two procedures are equivalent.
- It is dominated by the full-sample oracle, as expected for an upper bound with look-ahead.

Central conclusion. The out-of-sample relation is consistent with a stable, irregular cross-sectional pattern of benchmark-relative residual returns. Calling this "mispricing" would require additional economic identification. Discarding older data and re-estimating from a 120-month rolling window adds noise rather than dynamic forecasting information.

12.6 Minimal implementation pattern

```
for cutoff in cutoffs:
    future = future_alpha(cutoff, horizon=12, frozen_betas=True)
    rolling = rank_from_last_120_months(cutoff)
    expanding = rank_from_all_history_to(cutoff)

    rho_roll = spearmanr(rolling, future).statistic
    rho_expand = spearmanr(expanding, future).statistic
    increments.append(rho_roll - rho_expand)

estimate, se, p_one_sided = hac_mean_test(increments, alternative="greater")
```

Chapter 13

Residual diagnostics, benchmark sensitivity and model risk

13.1 Diagnostic tests

Test	Null hypothesis	If rejected
Jarque-Bera	Residual skewness and kurtosis match a normal distribution	Gaussian likelihood is questionable
Ljung-Box (lag 12)	No residual autocorrelation through lag 12	Serial dependence remains
ARCH-LM (lag 12)	No conditional heteroskedasticity	Residual variance is time-varying

For residuals $e_1,\ldots,e_T$, let S denote sample skewness and K sample kurtosis. The Jarque-Bera statistic is

$$JB = T/6 \cdot [S^2 + (K-3)^2/4] \approx \chi^2_2 \quad \text{under an IID normal null.} \tag{13.1}$$

Let $\hat{\rho}_\ell$ be the residual autocorrelation at lag ℓ. The Ljung-Box portmanteau statistic through lag m is

$$Q(m) = T(T+2)\sum_{\ell=1}^{m} \hat{\rho}^2_\ell/(T-\ell) \approx \chi^2_m. \tag{13.2}$$

For ARCH-LM, run the auxiliary regression

$$\hat{e}^2_t = c + \gamma_1 \hat{e}^2_{t-1} + \cdots + \gamma_m \hat{e}^2_{t-m} + \eta_t. \tag{13.3}$$

Under $H_0{:}\gamma_1{=}\cdots{=}\gamma_m{=}0$, the statistic $(T-m)R^2$ is asymptotically $\chi^2{}_m$. Rejection indicates predictable conditional variance, not necessarily incorrect conditional mean.

```
from statsmodels.stats.diagnostic import acorr_ljungbox, het_arch
from statsmodels.stats.stattools import jarque_bera

jb_stat, jb_p, skew, kurt = jarque_bera(resid)
lb_p = acorr_ljungbox(resid, lags=[12], return_df=True)["lb_pvalue"].iloc[0]
arch_stat, arch_p, _, _ = het_arch(resid, nlags=12)
```

In the paper, normality and no-ARCH are rejected for all 25 portfolios, while lag-12 independence is rejected for 16. This explains why IID-normal GRS is secondary, but it does not mean every conclusion is invalid: HAC and block resampling target specific inferential problems.

Question	What diagnostics imply
Are OLS coefficients algebraically defined?	Yes, provided X has full column rank.
Are OLS coefficients consistent for the linear projection?	Potentially, under orthogonality and suitable weak-dependence conditions; normality is not required.
Are conventional IID standard errors valid?	No when heteroskedasticity or serial dependence is material.
Does HAC prove the factor model is correctly specified?	No. It repairs an asymptotic covariance estimate, not omitted variables, nonlinearities or unstable betas.
Is a Gaussian SFA likelihood trustworthy after these rejections?	Only as a deliberately narrow diagnostic, reinforced by simulation and boundary/FDR safeguards.

13.2 FF3 versus FF5

FF5 adds profitability (RMW) and investment (CMA). Compare models only on their common sample. Otherwise, “model difference” is confounded with “sample difference.”

- Fit both models on July 1963-November 2025.
- Compare posterior ranks, score correlations, FDR discoveries and joint tests.
- Report the largest absolute rank move and the affected portfolio.

Paper result. FF3 and FF5 ranks correlate 0.875, yet ME5 BM2 moves from rank 9 to rank 20. Benchmark choice is economically meaningful even when overall rank correlation looks high.

13.3 Window-length sensitivity

Compare 60-, 120- and 180-month estimates on aligned dates. Rank correlation measures ordering agreement; Jaccard measures top/bottom membership agreement. Window length is a modelling decision, not a neutral knob.

Chapter 14

Stochastic frontier analysis: full derivation and its narrow role here

Derive the half-normal composite-error density, conditional distribution, conditional moments, likelihood and score; understand the boundary test, unit invariance, identification limits and numerical implementation.

14.1 From a production frontier to a portfolio-residual diagnostic

The classical production-frontier model says observed log output cannot exceed a stochastic frontier except through symmetric noise:

$$y_t = {x_t}'\beta + v_t - u_t, v_t \sim N(0, \sigma_v^2), u_t \sim N^+(0, \sigma_u^2), u_t \perp v_t. \tag{14.1}$$

$N^+(0,{\sigma_u}^2)$ denotes a zero-mean normal distribution truncated below at zero, equivalently $|N(0,{\sigma_u}^2)|$. Define the composite disturbance

$$\varepsilon_t = y_t - {x_t}'\beta = v_t - u_t. \tag{14.2}$$

The symmetric component v represents ordinary noise; u is a non-negative shortfall. For a cost frontier the sign is reversed, ε=v+u, because inefficiency raises cost. Sign orientation must be specified before estimation.

In this portfolio application, ε is a factor-model residual and u is merely a candidate one-sided residual component. Calling it managerial inefficiency would over-interpret the model: omitted factors, crash exposure, nonlinear payoffs, stale prices and changing volatility can also create negative skewness.

14.2 Component densities and change of variables

Let ϕ and Φ be the standard normal density and cumulative distribution function:

$$\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}, \Phi(z) = \int_{-\infty}^{z} \phi(s) ds. \tag{14.3}$$

The component densities are

$$f_v(v) = \sigma_v^{-1}\phi(v/\sigma_v), \tag{14.4}$$

$$f_u(u) = 2\sigma_u^{-1}\phi(u/\sigma_u) \cdot 1u \geq 0. \tag{14.5}$$

Because ε=v−u, we have v=ε+u. The transformation from (v,u) to (ε,u) has Jacobian determinant one. Independence therefore gives the joint density

$$f_{\varepsilon,u}(\varepsilon, u) = f_v(\varepsilon + u)f_u(u), u \geq 0. \tag{14.6}$$

Substitution yields

$$f_{\varepsilon,u}(\varepsilon, u) = [1/(\pi\sigma_v\sigma_u)]exp{-}\frac{1}{2}[(\varepsilon + u)^2/\sigma_v^2 + u^2/\sigma_u^2] \cdot 1u \geq 0. \tag{14.7}$$

14.3 Complete the square

Introduce total scale σ and shape ratio λ:

$$\sigma^2 = \sigma_v^2 + \sigma_u^2, \lambda = \sigma_u/\sigma_v. \tag{14.8}$$

Define the conditional-location and conditional-scale quantities

$$\mu_* = -\varepsilon\sigma_u^2/\sigma^2, \sigma_*^2 = \sigma_u^2\sigma_v^2/\sigma^2. \tag{14.9}$$

Now expand the left side and verify the key identity:

$$(\varepsilon + u)^2/\sigma_v^2 + u^2/\sigma_u^2 = (u - \mu_*)^2/\sigma_*^2 + \varepsilon^2/\sigma^2. \tag{14.10}$$

To see why, the coefficient on u^2 is

$$1/\sigma_v^2 + 1/\sigma_u^2 = \sigma^2/(\sigma_u^2\sigma_v^2) = 1/\sigma_*^2, \tag{14.11}$$

and the coefficient on u is

$$-2\mu_*/\sigma_*^2 = 2\varepsilon/\sigma_v^2. \tag{14.12}$$

The remaining constant also agrees because ${\mu_*}^2/{\sigma_*}^2$+ε^2/σ^2=$\varepsilon^2/{\sigma_\mathrm{v}}^2$. This single square-completion step produces both the marginal likelihood and the conditional distribution.

14.4 Deriving the composite-error density

Integrate u out of the joint density:

$$f_\varepsilon(\varepsilon) = \int_0^\infty f_{\varepsilon,u}(\varepsilon, u)du. \tag{14.13}$$

After the square completion, the u-integral is the upper probability of a N(μ_*,σ_*^2) variable:

$$\int_0^\infty exp[-(u-\mu_*)^2/(2\sigma_*^2)]du = \sigma_*\sqrt{2\pi}\Phi(\mu_*/\sigma_*). \tag{14.14}$$

Since $\sigma_*/(\sigma_u\sigma_v)=1/\sigma$ and

$$\mu_*/\sigma_* = -\varepsilon\sigma_u/(\sigma\sigma_v) = -\lambda\varepsilon/\sigma, \tag{14.15}$$

the half-normal SFA composite-error density is

$$f_\varepsilon(\varepsilon) = (2/\sigma)\phi(\varepsilon/\sigma)\Phi(-\lambda\varepsilon/\sigma). \tag{14.16}$$

When $\lambda=0$, $\Phi(0)=1/2$ and the density reduces to $\sigma^{-1}\phi(\varepsilon/\sigma)$, the ordinary Gaussian error. Positive λ tilts mass toward negative residuals.

14.5 Deriving u conditional on the observed residual

Bayes' rule gives $f(u|\varepsilon)=f(\varepsilon,u)/f(\varepsilon)$. Cancelling all terms independent of u leaves a normal kernel restricted to $u\geq0$:

$$f(u|\varepsilon) = \phi[(u-\mu_*)/\sigma_*]/[\sigma_*\Phi(\mu_*/\sigma_*)], u \geq 0. \tag{14.17}$$

Thus

$$u|\varepsilon \sim N(\mu_*, \sigma_*^2) \quad \text{truncated below at 0.} \tag{14.18}$$

Let $z_*=\mu_*/\sigma_*$ and define the inverse Mills ratio $r(z)=\phi(z)/\Phi(z)$. Standard truncated-normal integration gives

$$E(u|\varepsilon) = \mu_* + \sigma_* r(z_*), \tag{14.19}$$

$$\text{mode}(u|\varepsilon) = max(0, \mu_*), \tag{14.20}$$

$$Var(u|\varepsilon) = \sigma_*^2[1 - z_* r(z_*) - r(z_*)^2]. \tag{14.21}$$

The first expression is the Jondrow-Lovell-Materov-Schmidt conditional mean. A very negative ε makes μ_* positive, shifting conditional mass toward a larger one-sided shortfall. A positive ε makes μ_* negative, but truncation ensures $E(u|\varepsilon)$ remains non-negative.

14.6 Deriving the conditional exponential score

In production SFA, exp(−u) is often interpreted as technical efficiency when y is log output. Here returns have arbitrary reporting units, so first derive the more general moment for c>0:

$$E[e^{-cu}|\varepsilon] = [1/\Phi(\mu_*/\sigma_*)] \int_0^\infty e^{-cu}\phi[(u-\mu_*)/\sigma_*]du/\sigma_*. \tag{14.22}$$

Complete a second square in the exponent:

$$-cu - (u-\mu_*)^2/(2\sigma_*^2) = -[u-(\mu_* - c\sigma_*^2)]^2/(2\sigma_*^2) - c\mu_* + c^2\sigma_*^2/2. \tag{14.23}$$

The remaining integral is another truncated normal probability, giving

$$E[e^{-cu}|\varepsilon] = exp(-c\mu_* + c^2\sigma_*^2/2) \cdot \Phi[(\mu_* - c\sigma_*^2)/\sigma_*]/\Phi(\mu_*/\sigma_*). \tag{14.24}$$

For classical log-output efficiency set c=1. For a return series, use c=1/σ and define the scale-free diagnostic score

$$TE_{scale-free}(\varepsilon) = E[exp(-u/\sigma)|\varepsilon] \tag{14.25}$$

$$= exp[-\mu_*/\sigma + \frac{1}{2}(\sigma_*/\sigma)^2] \cdot \Phi[\mu_*/\sigma_* - \sigma_*/\sigma]/\Phi(\mu_*/\sigma_*). \tag{14.26}$$

Why is this invariant? If returns are multiplied by a positive constant a—for example, decimals to percentages—then ε, σ, $\mu*$ and $\sigma*$ all multiply by a. Every ratio in the last expression is unchanged. In contrast, exp(−u) changes under a unit conversion and is unsuitable for ranking return series reported on different scales.

14.7 A numerical conditional-shortfall example

Take σ_{v}=0.015, σ_{u}=0.010 and an observed residual ε=−0.020. Then

$$\sigma = 0.01803, \lambda = 0.6667, \mu_* = 0.006154, \sigma_* = 0.008321, z_* = 0.7396. \tag{14.27}$$

Because r(0.7396)=0.3940,

$$E(u|\varepsilon) = 0.006154 + 0.008321(0.3940) = 0.009432. \tag{14.28}$$

The conditional mode is 0.006154 and the scale-free exponential score is approximately 0.6257. If all returns are multiplied by 100, recomputation produces exactly the same 0.6257 score up to numerical rounding.

14.8 Log-likelihood and analytic score

For observation t, define $z_t = \varepsilon_t/\sigma$ and $h_t = -\lambda z_t$. The log-likelihood contribution is

$$\ell_t(\beta, \sigma, \lambda) = log2 - log\sigma + log\phi(z_t) + log\Phi(h_t). \tag{14.29}$$

The sample log-likelihood is $\ell = \sum_{t=1}^{T} \ell_t$. Let $R(h) = \phi(h)/\Phi(h)$. Holding λ fixed, differentiation gives

$$\partial \ell_t/\partial \varepsilon_t = -[z_t + \lambda R(h_t)]/\sigma. \tag{14.30}$$

Since $\varepsilon_t = y_t - x_t{}'\beta$ and $\partial\varepsilon_t/\partial\beta = -x_t$,

$$\partial \ell_t/\partial \beta = x_t[z_t + \lambda R(h_t)]/\sigma. \tag{14.31}$$

Holding ε and λ fixed, $\partial z/\partial\sigma = -z/\sigma$ and $\partial h/\partial\sigma = \lambda z/\sigma$, so

$$\partial \ell_t/\partial \sigma = [-1 + z_t^2 + \lambda z_t R(h_t)]/\sigma. \tag{14.32}$$

Finally, holding ε and σ fixed, $\partial h/\partial\lambda = -z$, hence

$$\partial \ell_t/\partial \lambda = -z_t R(h_t). \tag{14.33}$$

If optimisation uses $\eta = \log\ \sigma$ and $\kappa = \log\ \lambda$, apply the chain rule: $\partial\ell/\partial\eta = \sigma\partial\ell/\partial\sigma$ and $\partial\ell/\partial\kappa = \lambda\partial\ell/\partial\lambda$. A pure log λ parameterisation cannot represent $\lambda = 0$, so estimate the Gaussian null separately with σ_u fixed at zero.

14.9 Numerical implementation that can survive difficult tails

```
from scipy.special import log_ndtr
from scipy.stats import norm

def loglik(theta, y, X):
    beta = theta[:-2]
    sigma = np.exp(theta[-2])       # positive total scale
    lam = np.exp(theta[-1])         # positive under alternative
    eps = y - X @ beta
    z = eps / sigma
    return np.sum(np.log(2.0) - np.log(sigma)
                 + norm.logpdf(z) + log_ndtr(-lam * z))

# Fit Gaussian null separately: lambda = sigma_u = 0.
# Check analytic score against central finite differences.
# Use multiple starts and record optimiser convergence/gradient norm.
```

- Use log Φ via `log_ndtr`; directly evaluating Φ in a far tail can underflow to zero.
- Evaluate inverse Mills ratios with stable log-density minus log-CDF calculations.
- Parameterise positive scales logarithmically and transform back only for reporting.
- Check the analytic gradient against a high-accuracy central numerical gradient at random interior parameter values.
- Simulate known parameters, refit, and verify bias, coverage, null rejection and unit invariance.

14.10 Boundary likelihood-ratio testing

The null H_0:σ_u=0 is on the boundary $\sigma_u \geq 0$. Under the null, λ=0 and the model collapses to a Gaussian disturbance. Let ℓ_1 be the maximised SFA log-likelihood and ℓ_0 the maximised Gaussian-null log-likelihood:

$$LR = 2(\ell_1 - \ell_0) \geq 0. \tag{14.34}$$

The usual $\chi^2{}_1$ reference assumes an interior null and is invalid. Under standard boundary regularity conditions,

$$LR \Rightarrow \frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2, \tag{14.35}$$

where $\chi^2{}_0$ is a point mass at zero. Therefore

$$p = 1 \text{ if } LR = 0; p = \frac{1}{2} \cdot P(\chi_1^2 \geq LR) \text{ if } LR > 0. \tag{14.36}$$

There are 25 portfolio-specific SFA boundary tests, so apply BH FDR to those 25 p-values as one prespecified family. Only portfolios with q≤0.05 receive an `sfa_rank`; all others are explicitly marked unsupported rather than quietly ordered.

14.11 Identification: why shape, not merely location, matters

For the half-normal distribution, E(u)=$\sigma_u\sqrt{2/\pi}$, so

$$E(\varepsilon) = E(v - u) = -\sigma_u\sqrt{2/\pi}. \tag{14.37}$$

A regression intercept and a non-zero mean shortfall can both explain a constant downward location shift. If the model contains an unrestricted intercept, separating that intercept from E(u) relies primarily on distributional shape—especially skewness—not on the mean alone. That is weak evidence when residuals are non-normal, serially dependent and ARCH.

More formally, identification asks whether distinct parameter vectors generate distinct observable distributions. If a flexible location term, omitted factors or time-varying volatility can mimic the skewness attributed to u, the economic label attached to σ_u is not securely identified even when the optimiser returns a precise number.

Interpretation boundary. A rejected Gaussian boundary null supports one-sided residual asymmetry under this parametric model. It does not identify managerial skill, technical efficiency, arbitrage or mispricing. The factor alpha remains the paper's primary performance estimand.

14.12 Simulation verification plan

Experiment	Data-generating process	Expected result
Gaussian null	σ_u=0	Boundary-test rejection near its nominal level after correct mixture calibration

Experiment	Data-generating process	Expected result
Interior SFA	$\sigma_u > 0$ with known $\beta, \sigma_v, \sigma_u$	Parameter bias falls with T; conditional-score code matches numerical integration
Unit rescaling	Multiply y and $X\beta$ by 100	β/scale transforms appropriately; scale-free score and ranks stay unchanged
Misspecified skewness	Skewed or jump residuals without half-normal u	SFA may reject, demonstrating that rejection is not unique evidence of inefficiency
ARCH residuals	Gaussian mean with time-varying variance	Assess false one-sided detections caused by volatility dynamics

```
for replication in range(R):
    v = rng.normal(0, sigma_v, T)
    u = np.abs(rng.normal(0, sigma_u, T))
    y = X @ beta + v - u
    fitted = fit_half_normal_sfa(y, X)
    store(fitted.params, fitted.lr_pvalue, fitted.score)

assert gradient_error < tolerance
assert max_abs(score_decimal - score_percent) < 1e-10
```

Paper result. Only ME5 BM4 survives the 25-test SFA boundary BH correction: raw p=0.000254 and q=0.006361. Therefore, the evidence supports one narrow asymmetry diagnostic, not a general 25-portfolio efficiency league table.

Chapter 15

External validation, source reconciliation and status semantics

15.1 Four different validation questions

Validation	Question	Evidence
Integrity	Is the local analytical panel internally valid?	Keys, nulls, calendar, balance, factor consistency
Numerical replication	Does an independent implementation reproduce calculations?	statsmodels coefficient/HAC match
Source reconciliation	Do local bytes match an identifiable official release?	Hashes, archives, overlapping-cell comparison
Literature anchor	Are broad patterns coherent with established evidence?	5 × 5 loadings/returns; historical sample checks

Do not collapse these into "validated." A calculation can reproduce perfectly on a locally corrupted input. Conversely, a fixed historical vintage can be internally valid even when it differs from a later revised official snapshot.

15.2 Status vocabulary

Status	Meaning	Blocks release?
PASS	The stated valid check completed successfully	No
FAIL	A required and valid check failed	Yes
UNRESOLVED	A discrepancy exists but its cause cannot be determined	Visible; policy-dependent
NOT COMPARABLE	The sources/vintages do not support the proposed equality comparison	No, but must be disclosed

15.3 Known-result validation without result fitting

Check broad expectations—such as HML loading increasing toward high book-to-market and SMB exposure changing across size groups—but treat them as diagnostics. Unexpected patterns trigger investigation; they must not be overwritten or tuned away.

```
comparison = local.merge(official, on=["date","portfolio"],
↪  suffixes=("_local","_official"))
comparison["abs_diff"] = (
    comparison["return_local"] - comparison["return_official"]
).abs()
summary = {
    "max_abs_diff": comparison["abs_diff"].max(),
    "date_of_max": comparison.loc[comparison["abs_diff"].idxmax(), "date"],
    "n_above_tol": int((comparison["abs_diff"] > tolerance).sum()),
}
```

Chapter 16

Complete reproducible implementation architecture

16.1 Suggested repository structure

```
project/
|-- analysis/
|   |-- data.py                    # parsing, joins, integrity
|   |-- factor_models.py           # OLS and frozen-beta evaluation
|   |-- joint_hac.py               # full alpha covariance
|   |-- empirical_bayes.py         # profile ML and posterior
|   |-- multiple_testing.py        # BH families
|   |-- bootstrap.py               # common-date circular blocks
|   |-- persistence.py             # overlap and transitions
|   |-- forward_validation.py      # strict forward tests
|   |-- incremental_validation.py  # rolling vs benchmark rankings
|   |-- diagnostics.py             # JB, LB, ARCH, SFA
|   |-- external_validation.py     # source and statsmodels checks
|   `-- run_all.py                 # canonical orchestration
|-- config/
|   `-- analysis.yaml
|-- data/
|   |-- raw/                       # immutable
|   `-- SOURCES.md
|-- results/
|   |-- tables/
|   |-- figures/
|   `-- run_manifest.json
|-- reports/
|   `-- generate_report.py
|-- tests/
|-- requirements-dev-lock.txt
`-- README.md
```

16.2 Canonical pipeline

Load + validate

Estimate + infer

Resample + validate

Export canonical tables

Generate report

```
python -m pip install -r requirements-dev-lock.txt
pytest -q
ruff check .
MPLBACKEND=Agg python -m analysis.run_all
python -m analysis.export_tables
python reports/generate_report.py
```

16.3 Run manifest

A run manifest should record:

- input paths, byte sizes and hashes;
- sample dates, dimensions and weighting block;
- model, factor columns, return units and annualisation;
- HAC lag/kernel/correction and covariance diagnostics;
- bootstrap type, block length, draws and seed;
- rolling window, step, forward horizon and comparison benchmarks;
- software/Python versions, Git commit if available, runtime and output hashes.

16.4 Essential tests

16.4.1 Data tests

- Duplicate key rejection
- Missing month rejection
- Panel balance
- Factor consistency
- Unit and block detection

16.4.2 Estimator tests

- OLS vs statsmodels
- Joint HAC diagonal special case
- PSD safeguards
- EB shrinkage simulations
- GRS/Wald null and alternative

16.4.3 Validation tests

- No-look-ahead dates
- Frozen-beta invariants
- Common-date bootstrap
- BH edge cases
- Increment sign conventions

16.4.4 Reporting tests

- Headline/table consistency
- No p-value printed as zero
- Status semantics
- All claims generated from canonical tables

- Readable rendered pages

16.5 Two notebooks, two purposes

A beginner benefits from separating conceptual verification from full empirical reproduction. The teaching notebook should be tiny enough to inspect every matrix. The replication notebook should call production functions rather than contain a second, drifting implementation.

Notebook	Suggested cells	Required outputs
01_mathematical_foundations.ipynb	Simulate factors and three portfolios; derive OLS by hand; compare matrix OLS; build Γ_0 and Γ_1; complete the EB square; generate circular blocks; simulate half-normal SFA; compare analytic/numerical gradients	Small matrices printed in full, hand-versus-code equality assertions, conditional-density integral equal to one, SFA unit-invariance assertion
02_full_replication.ipynb	Load manifest; call canonical data validation; fit FF3; joint HAC/Wald/GRS/BH; EB; bootstrap; persistence; forward/incremental tests; diagnostics/SFA; FF5 and source checks	Canonical tables loaded from disk, headline reconciliation table, hashes and run metadata; no duplicated estimator code

16.6 A literate cell pattern

```
# Question
# Does the full joint HAC diagonal reproduce the 25 separate HAC variances?

# Mathematical object
# V_HAC = c * [Gamma_0 + sum_l w_l(Gamma_l + Gamma_l.T)]

# Computation
V_joint = joint_newey_west_alpha_covariance(resid, X, lags=12)
v_separate = np.array([single_alpha_hac_variance(resid[:, i], X, 12)
                       for i in range(N)])

# Falsifiable check
np.testing.assert_allclose(np.diag(V_joint), v_separate, rtol=1e-10)

# Interpretation
# Equality validates implementation consistency; it does not validate the model.
```

Each notebook section should state the question, mathematical object, computation, falsifiable check and interpretation. This makes the notebook an audit trail rather than a scrapbook of commands.

Chapter 17

How to read the paper's evidence correctly

Finding	What it supports	What it does not support
Joint HAC/Wald p = 6.42e-10	Not all FF3 alphas are zero	Positive skill or stable ranking
Five raw alphas survive BH	Five model-relative intercept discoveries	Five positive winners; four are negative
Two posterior intervals wholly positive	Two pooled alphas remain positive under the EB model	Those portfolios are guaranteed to outperform
Wide bootstrap rank intervals	Point ranks are uncertain	Scores are useless; uncertainty is itself information
120-month rank persistence 0.156	Weak decade-to-decade structural persistence	Short-horizon 0.863 proves stability
Forward spread 243 bps/year	Past ordering has average future information	Annual rolling timing is useful
Rolling increment vs expanding = -0.042	Re-estimation adds no dynamic predictive power	No stable cross-sectional information exists
FF3/FF5 rank correlation 0.875; max move 11	Broad agreement plus material benchmark risk	Alpha is benchmark-independent
One SFA FDR discovery	One residual series shows supported one-sided asymmetry	A 25-portfolio efficiency league table

The disciplined final statement. There is cross-sectional FF3 alpha variation and a stable ordering with some forward information, but ranks are uncertain, dynamic rolling re-estimation adds no value over an expanding estimate, and conclusions depend on the benchmark and source vintage.

Chapter 18

Capstone: reproduce the complete study

Complete the following in order. Each box is stored locally in your browser when checked.

Document official data URLs, blocks, units, dates and hashes. Parse the value-weighted 25-portfolio returns and FF3 factors. Create one validated portfolio-month panel with a strict many-to-one factor join. Fit 25 FF3 regressions and reproduce coefficients independently. Construct the full 25 $\times$ 25 joint HAC alpha covariance. Run the joint HAC/Wald test and secondary GRS test. BH-correct the 25 raw-alpha tests as one family. Fit μ and τ and calculate multivariate EB posterior means and intervals. Run 5,000 full-refit common-date circular-block bootstrap draws. Report rank intervals and top/bottom quintile probabilities. Run 120-month rolling persistence at 12-, 60- and 120-month horizons with overlap labels. Build the non-overlapping quintile transition matrix and Jaccard measures. Run 89 strictly forward 12-month evaluations with frozen training estimates. Aggregate correlation and spread time series with HAC inference. Implement characteristic, expanding-window and oracle benchmark rankings. Test rolling-minus-benchmark increments with HAC and block bootstrap. Run FF3/FF5 common-sample and 60/120/180-month sensitivity analyses. Run Jarque-Bera, Ljung-Box and ARCH-LM residual diagnostics. Fit SFA only as a residual diagnostic; boundary-test and BH-correct its p-values. Separate local integrity, independent replication, source reconciliation and literature anchors. Write manifests, canonical CSVs, figures and a report with generated claims. Run tests/lint, render the report, and visually inspect every page.

18.1 Assessment standard

A successful capstone must not merely reproduce headline numbers. It must demonstrate no leakage, preserve dependence, define testing families, expose uncertainty, label non-comparability, and prevent unsupported interpretations.

Chapter 19

Exercises with guided solutions

Exercise 1 - Unit conversion

A monthly alpha is 0.00085. The linear annualised value is $0.00085 \times 12 \times 10{,}000 =$ **102 basis points/year**. This does not compound monthly returns.

Exercise 2 - Why is a many-to-many merge dangerous?

It replicates the same economic observation. Point estimates may be reweighted and standard errors treat duplicate rows as new information. Validate key uniqueness before and after every join.

Exercise 3 - Joint rejection with mostly uncertain individual ranks

A joint test aggregates evidence across the alpha vector using covariance. It can reject all-zero even while many individual posterior intervals cross zero and ranks overlap. Joint existence, individual magnitude and ordering precision are different questions.

Exercise 4 - BH calculation

With m = 25 and q = 0.05, the first two BH thresholds are 0.002 and 0.004. Raw p-values 0.000254 and 0.0063 produce only one discovery at these positions because the second exceeds 0.004, unless a later ordered p-value satisfies its larger threshold.

Exercise 5 - Why resample common dates?

If each portfolio resamples different dates, same-month common shocks are destroyed. Joint score covariance and rank co-movement are then understated.

Exercise 6 - Why refit μ and τ inside each bootstrap?

They are estimated quantities. Holding them fixed conditions away hyperparameter-estimation uncertainty and makes rank/score intervals too narrow.

Exercise 7 - Overlap arithmetic

Two 120-month windows 60 months apart share 60 months, so overlap is $(120-60)/120 = 50\%$. They are not independent persistence observations.

Exercise 8 - Frozen beta

Future alpha is evaluated relative to the risk exposure estimated from training data. Re-estimating beta on future observations lets evaluation data redefine the benchmark residual and introduces look-ahead.

Exercise 9 - Positive forward spread but no dynamic increment

A stable cross-sectional ordering can predict future outcomes. If the expanding-window ranking performs as well as the rolling rank, annual re-estimation contributes no additional timing information.

Exercise 10 - Why is the oracle not a fair competitor?

It uses the full sample, including evaluation periods, so it has look-ahead. It is an upper bound used to diagnose the strength of a fixed ordering, not an implementable strategy.

Exercise 11 - HAC versus GRS

GRS has an exact F reference under IID Gaussian errors. HAC/Wald is asymptotically robust to specified heteroskedasticity and autocorrelation. When diagnostics reject IID Gaussian residuals, GRS remains informative as a conventional reference but should not be primary.

Exercise 12 - Source mismatch

If local calculations reproduce exactly but local values differ from the current official snapshot, numerical implementation is validated while source equality is not. Report the two findings separately.

Exercise 13 - SFA interpretation

A supported one-sided residual component indicates asymmetry under the fitted distribution. It does not prove inefficiency, especially when omitted risks, nonlinearities and tail events can also create asymmetry.

Exercise 14 - Design a negative control

Simulate returns from the fitted factors with true alpha zero and dependent residuals. The pipeline should recover near-zero posterior centres, controlled false discoveries and no systematic forward increment. This tests calibration rather than reproducing the empirical result.

Exercise 15 - Complete the SFA square

Starting from $(\varepsilon+\mathrm{u})^2/\sigma_\mathrm{v}{}^2+\mathrm{u}^2/\sigma_\mathrm{u}{}^2$, collect coefficients of u^2 and u. Set $\sigma{*}^2=\sigma_\mathrm{u}{}^2\sigma_\mathrm{v}{}^2/(\sigma_\mathrm{u}{}^2+\sigma_\mathrm{v}{}^2)$ and $\mu{*}=-\varepsilon\sigma_\mathrm{u}{}^2/(\sigma_\mathrm{u}{}^2+\sigma_\mathrm{v}{}^2)$. Expanding $(\mathrm{u}-\mu{*})^2/\sigma{*}^2+\varepsilon^2/\sigma^2$ reproduces every coefficient, proving the identity used in both marginalisation and conditioning.

Exercise 16 - Derive the SFA marginal density

Insert the completed square into $\mathrm{f}(\varepsilon,\mathrm{u})$, factor out $\exp[-\varepsilon^2/(2\sigma^2)]$, and integrate the remaining $\mathrm{N}(\mu{*},\sigma{*}^2)$ kernel from zero to infinity. Use $\mu{*}/\sigma{*}=-\lambda\varepsilon/\sigma$ and $\sigma{*}/(\sigma_\mathrm{u}\sigma_\mathrm{v})=1/\sigma$ to obtain $\mathrm{f}(\varepsilon)=(2/\sigma)\phi(\varepsilon/\sigma)\Phi(-\lambda\varepsilon/\sigma)$.

Exercise 17 - Verify the SFA conditional mean

Standardise $\mathrm{w}=(\mathrm{u}-\mu{*})/\sigma{*}$. The lower bound becomes $-\mathrm{z}{*}$. Integration by parts gives $\int_{-\mathrm{z}*}{}^{\infty}\mathrm{w}\phi(\mathrm{w})\mathrm{dw}=\phi(\mathrm{z}{*})$. Divide by $\Phi(\mathrm{z}{*})$ and transform back: $\mathrm{E}(\mathrm{u}|\varepsilon)=\mu{*}+\sigma{*}\phi(\mathrm{z}{*})/\Phi(\mathrm{z}{*})$.

Exercise 18 - Prove unit invariance

Replace $\varepsilon,\sigma_\mathrm{u},\sigma_\mathrm{v}$ by a times themselves for a>0. Then $\sigma,\mu{*},\sigma{*}$ also multiply by a, while $\mu{*}/\sigma$, $\sigma{*}/\sigma$ and $\mu{*}/\sigma{*}$ do not change. Hence $\mathrm{E}[\exp(-\mathrm{u}/\sigma)|\varepsilon]$ is invariant. $\mathrm{E}[\exp(-\mathrm{u})|\varepsilon]$ is not.

Exercise 19 - Boundary p-value

If LR=6.635, the ordinary $\chi^2{}_1$ upper tail is approximately 0.010. The 50:50 boundary-mixture p-value is approximately 0.005. If LR=0, the point mass implies p=1. Apply BH to the resulting family of 25 SFA p-values.

Exercise 20 - Posterior shrinkage

With μ=-26, τ=94, raw alpha a=250 and standard error s=150 bps/year, the diagonal-model weight is $94^2/(94^2+150^2)$=0.282. The posterior mean is 0.282(250)+0.718(-26)=51.8 bps/year. The full-covariance answer need not equal this because other portfolios enter through off-diagonal sampling covariance.

Chapter 20

Terminology glossary

Alpha
Regression intercept: average return unexplained by the included linear factors.

Beta
Factor loading measuring conditional sensitivity to a factor.

Excess return
Portfolio return minus the risk-free return.

FF3
Fama-French market, size and value factor model.

FF5
FF3 augmented with profitability and investment factors.

Mkt-RF
Market portfolio return minus the risk-free rate.

SMB
Small-minus-big factor associated with firm size.

HML
High-minus-low book-to-market value factor.

RMW
Robust-minus-weak profitability factor.

CMA
Conservative-minus-aggressive investment factor.

Basis point
One hundredth of one percentage point; 100 bps = 1%.

Balanced panel
Every portfolio has an observation at every included date.

Analytical grain
The entity-time key represented by one row; here, one portfolio-month.

Provenance
Documented origin, date, version, transformation and identity of data.

SHA-256
Cryptographic digest used to identify exact file bytes.

OLS
Ordinary least squares regression.

Self-financing portfolio
A long-short portfolio whose long position is financed by its short position, so net initial investment is zero in the idealised construction.

Residual
Observed outcome minus fitted model outcome.

Influence weight
Coefficient showing how one observation's disturbance contributes linearly to an estimator such as alpha.

Heteroskedasticity
Non-constant conditional residual variance.

Autocorrelation
Dependence between observations at different times.

Cross-sectional dependence
Same-date dependence among different portfolios.

HAC
Heteroskedasticity- and autocorrelation-consistent covariance estimation.

Newey-West
A kernel-weighted HAC covariance estimator.

Long-run covariance
Sum of contemporaneous and lagged covariance matrices governing the asymptotic variance of a dependent sample mean.

Bartlett kernel
Linearly declining weights applied to lagged covariance terms.

Lag
Time separation between observations in dependence calculations.

Positive semi-definite
A covariance property ensuring non-negative variance in every direction.

Condition number
Numerical sensitivity of a matrix solve to perturbations.

Wald test
Quadratic-form test of parameter restrictions using an estimated covariance.

GRS test
Classical joint test that all factor-model alphas are zero under IID Gaussian assumptions.

Null hypothesis
The restriction evaluated by a statistical test.

p-value
Tail probability of evidence at least as extreme under the null model.

False discovery rate
Expected proportion of false rejections among all rejections.

BH procedure
Benjamini-Hochberg step-up multiple-testing correction.

q-value
Multiplicity-adjusted evidence associated with FDR control.

Empirical Bayes
Bayesian-form shrinkage with prior hyperparameters estimated from the observed ensemble.

Prior
Distribution describing the cross-sectional alpha population before each noisy estimate is combined with it.

Posterior
Updated distribution after combining prior structure with data.

Hyperparameter
Parameter of the prior distribution, such as μ or τ.

Marginal likelihood
Likelihood after integrating out latent parameters; used here to estimate μ and τ.

Profile likelihood
Likelihood in one parameter after replacing nuisance parameters by their conditional maximisers.

Partial pooling
Shrinking unit estimates toward a shared centre without forcing them to be equal.

Winner's curse
Upward noise bias in an estimate selected because it is the largest.

Block bootstrap
Resampling contiguous observations to preserve local time dependence.

Circular block
A block allowed to wrap from the sample end to the beginning.

Common-date resampling
Applying identical resampled dates to every portfolio.

Rank interval
Empirical uncertainty range for an item's rank.

Rolling window
Fixed-length historical sample that moves through time.

Expanding window
Historical sample that retains all available data up to each cutoff.

Look-ahead
Use of information unavailable at the decision time.

Leakage
Any path by which evaluation information influences training, preprocessing or selection.

Frozen beta
A factor loading fixed at the training cutoff during future evaluation.

Spearman correlation
Correlation of ranks; a measure of monotone ordering agreement.

Fisher transformation
atanh transformation used when aggregating correlations.

Quintile spread
Mean outcome of the top fifth minus mean outcome of the bottom fifth.

Oracle
Look-ahead benchmark used as an upper bound, not an implementable strategy.

Jaccard index
Set overlap measured as intersection divided by union.

Persistence
Stability of scores or ranks across time.

Mobility
Movement between rank or quintile states over time.

Jarque-Bera
Residual normality diagnostic based on skewness and kurtosis.

Ljung-Box
Portmanteau test for serial autocorrelation.

ARCH-LM
Test for autoregressive conditional heteroskedasticity.

SFA
Stochastic frontier analysis separating symmetric noise and one-sided shortfall.

Composite error
Disturbance formed from multiple latent components; in production SFA, ε=v−u.

Half-normal
Distribution of the absolute value of a zero-mean normal variable.

Truncated normal
A normal distribution conditioned to lie inside a specified interval.

Inverse Mills ratio
Ratio ϕ(z)/Φ(z), appearing in moments of a one-sided truncated normal.

Likelihood
Observed-data density viewed as a function of unknown parameters.

Maximum likelihood
Parameter estimation by maximising the sample likelihood or log-likelihood.

Score
Gradient of the log-likelihood with respect to its parameters; distinct here from a portfolio performance score.

Mixture distribution
Probability distribution formed by weighting component distributions; the SFA boundary reference mixes ${\chi^2}_0$ and ${\chi^2}_1$.

Identification
Property that distinct parameter values imply distinct observable probability distributions.

Boundary test
Test where the null parameter lies on the edge of its permitted space.

Model risk
Conclusion sensitivity to benchmark, assumptions, specification or tuning.

Manifest
Machine-readable record of inputs, settings, environment and outputs.

NOT COMPARABLE
Status indicating that exact equality is not a valid check for the available versions.

Chapter 21

Primary references and further reading

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These notes paraphrase and teach the methods used in the supplied working paper. Numerical results are study-specific and should be regenerated from the paper's canonical data and code rather than copied into another analysis.